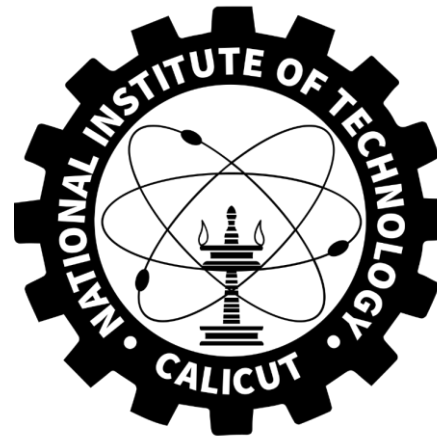
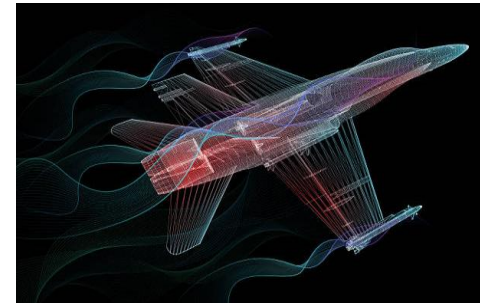


ME4028D/AERODYNAMICS

Faculty

Dr. Sanjeev K Manjhi

Assistant Professor



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Syllabus and Books

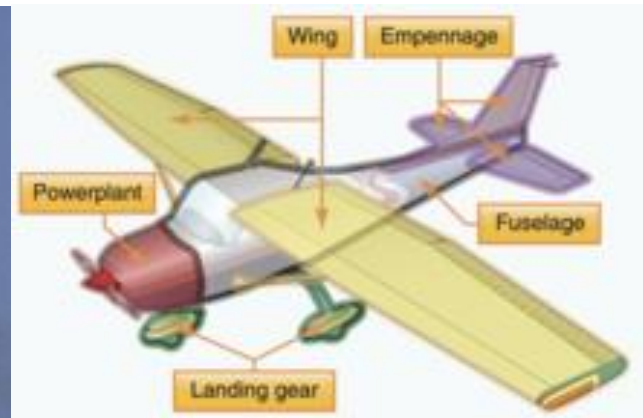
- (i) Basic fluid flow equations, Reynolds transport theorem, integral and differential formulations of continuity and momentum equations, Navier-Stokes equation; equations for incompressible inviscid flows; fluid circulation and rotation, vorticity; stream function; velocity potential; complex potential; Blasius theorem for force and moment on bodies; governing equations for irrotational incompressible flow; elementary flow patterns and their superposition.
- (ii) Flow past a cylinder: lifting flow and non-lifting flow; Magnus effect; Kutta-Joukowski theorem; conformal transformation: Joukowski transformation; airfoils: nomenclature; Kelvin's circulation theorem; vortex sheet; Kutta condition; thin airfoil theory: symmetric airfoil, cambered airfoil; airfoil characteristics: flow over airfoil; comparison with the real case; aerodynamic center, pitching moment.
- (iii) Incompressible flow over finite wings: downwash and induced drag; vortex filament; Biot-Savart Law; Helmholtz theorems; horseshoe vortex; ground effects; Prandtl's lifting line theory: elliptic lift distribution, general lift distribution; effect of aspect ratio: swept and delta wings; airfoils in compressible flow; small-perturbation theory.

References:

1. **J. D. Anderson, Fundamentals of Aerodynamics, 5th ed. McGraw Hill, 2010.**
2. E. L. Houghton, P. W. Carpenter, S. H. Collicott, and D. T. Valentine, Aerodynamics for Engineering Students, 6th ed. Butterworth-Heinemann, 2013.
3. A. M. Kuethe and C. Chow, Foundations of Aerodynamics: Bases of Aerodynamic Design, 5th ed. John Wiley and Sons, 1998.
4. L. Katz and A. Plotkin, Low Speed Aerodynamics. Cambridge University Press, 2001.
5. **Ronald L. Panton, Incompressible Flo, 4th ed. John Wiley & Sons, 2013.**



Wright Brothers, 1903





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NACA 4421



NACA 0030



NACA 4412



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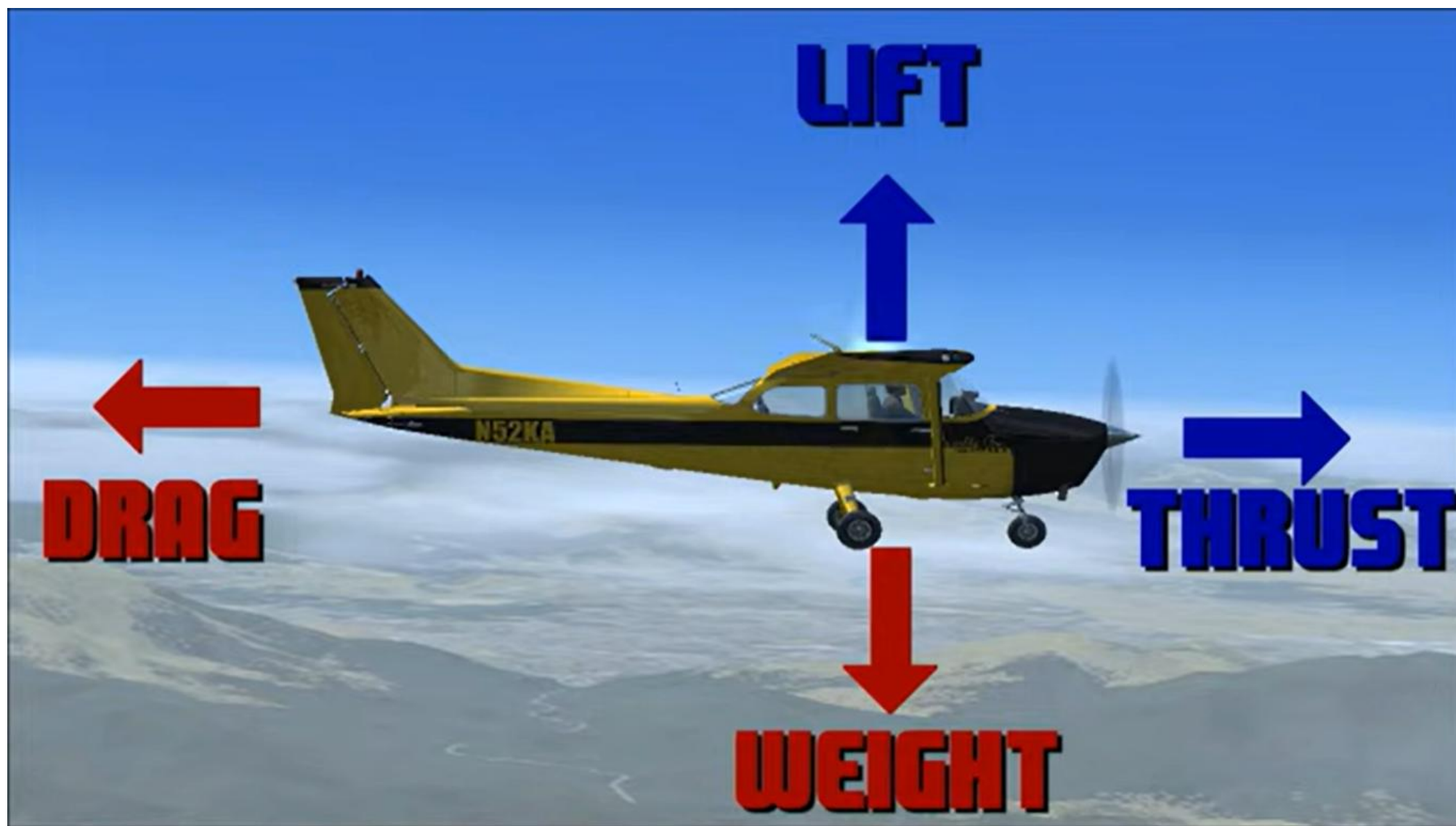
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DOUBLE WEDGE



Aerodynamics

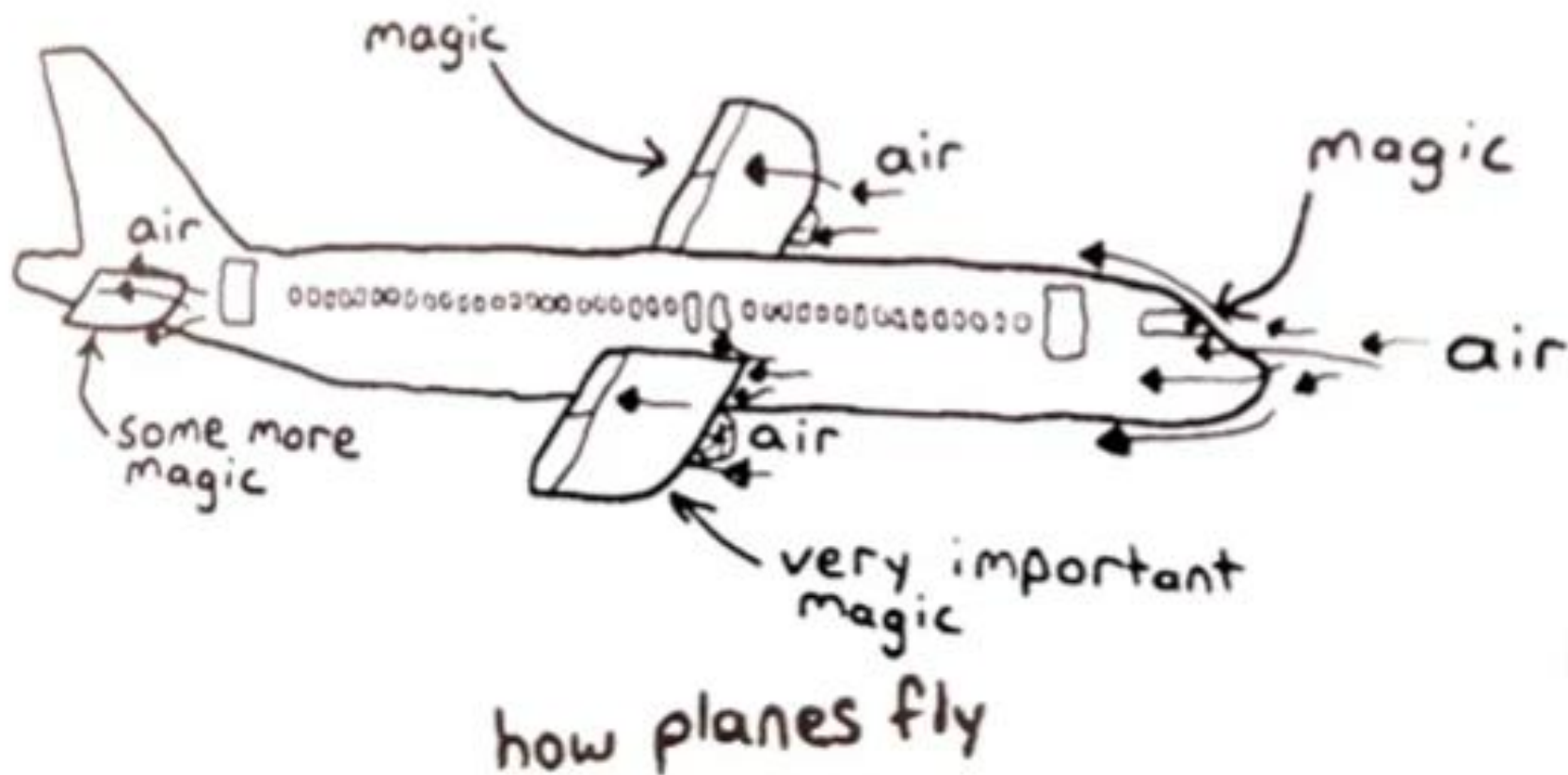
Some Introductory Thoughts

Incompressible

1. Some Fundamental Principles and Equations
 2. Fundamentals of Inviscid, Incompressible Flow
 3. Incompressible Flows over Airfoils
 4. Incompressible Flow over Finite Wings
 5. Three-Dimensional Incompressible Flow
-

Introduction to Aerodynamics

Some Introductory Thoughts



- Some Introductory Thoughts

The term “aerodynamics” is generally used for problems arising from flight and other topics involving the flow of air.

Ludwig Prandtl, 1949

Aerodynamics: The dynamics of gases, especially atmospheric interactions with moving objects.

**The American Heritage
Dictionary of the English**

Language. 1969



Douglas DC-3



Boeing 707



Bell X-1



Lockheed F-104



F-22 Raptor



Blended wing body

Rectilinear stream of discrete particles

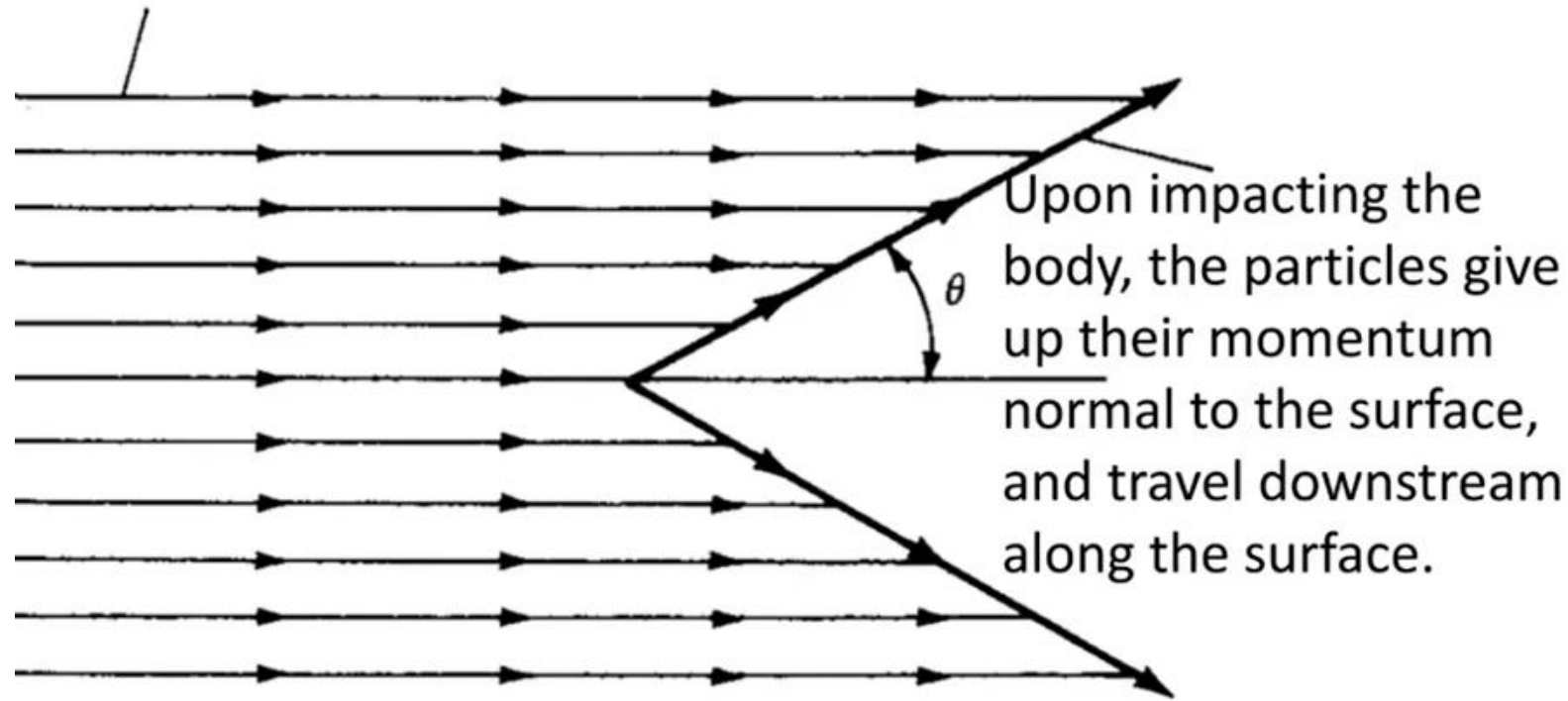
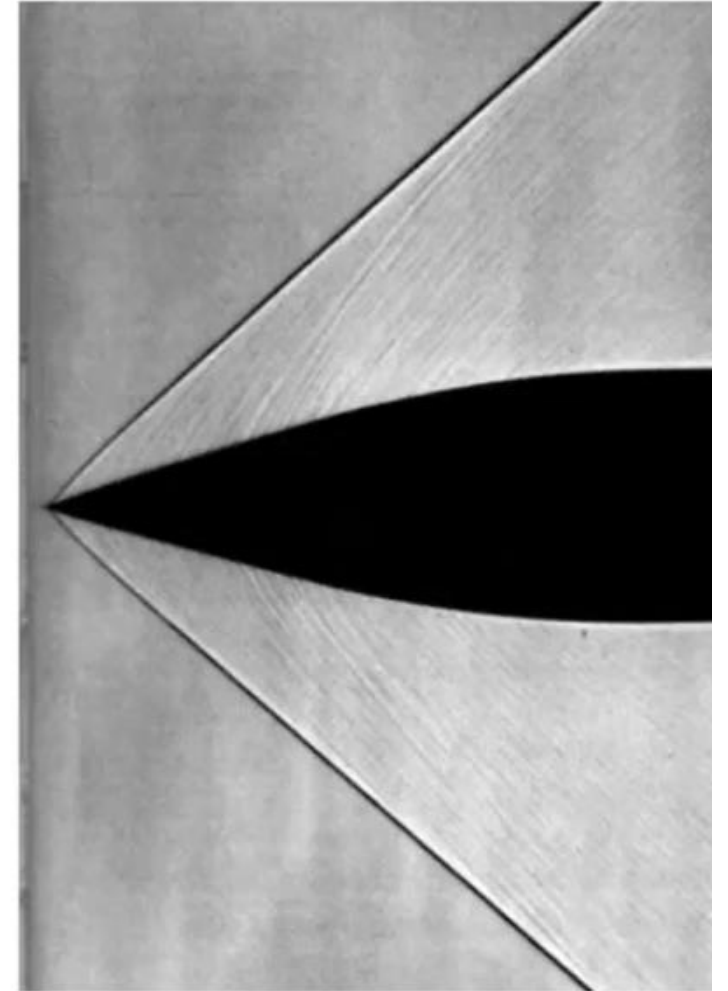


Figure 1.1 Isaac Newton's model of fluid flow in the year 1687.

This model was widely adopted in the seventeenth and eighteenth centuries but was later found to be conceptually inaccurate for most fluid flows.





1901 Wind Tunnel

Glenn
Research
Center

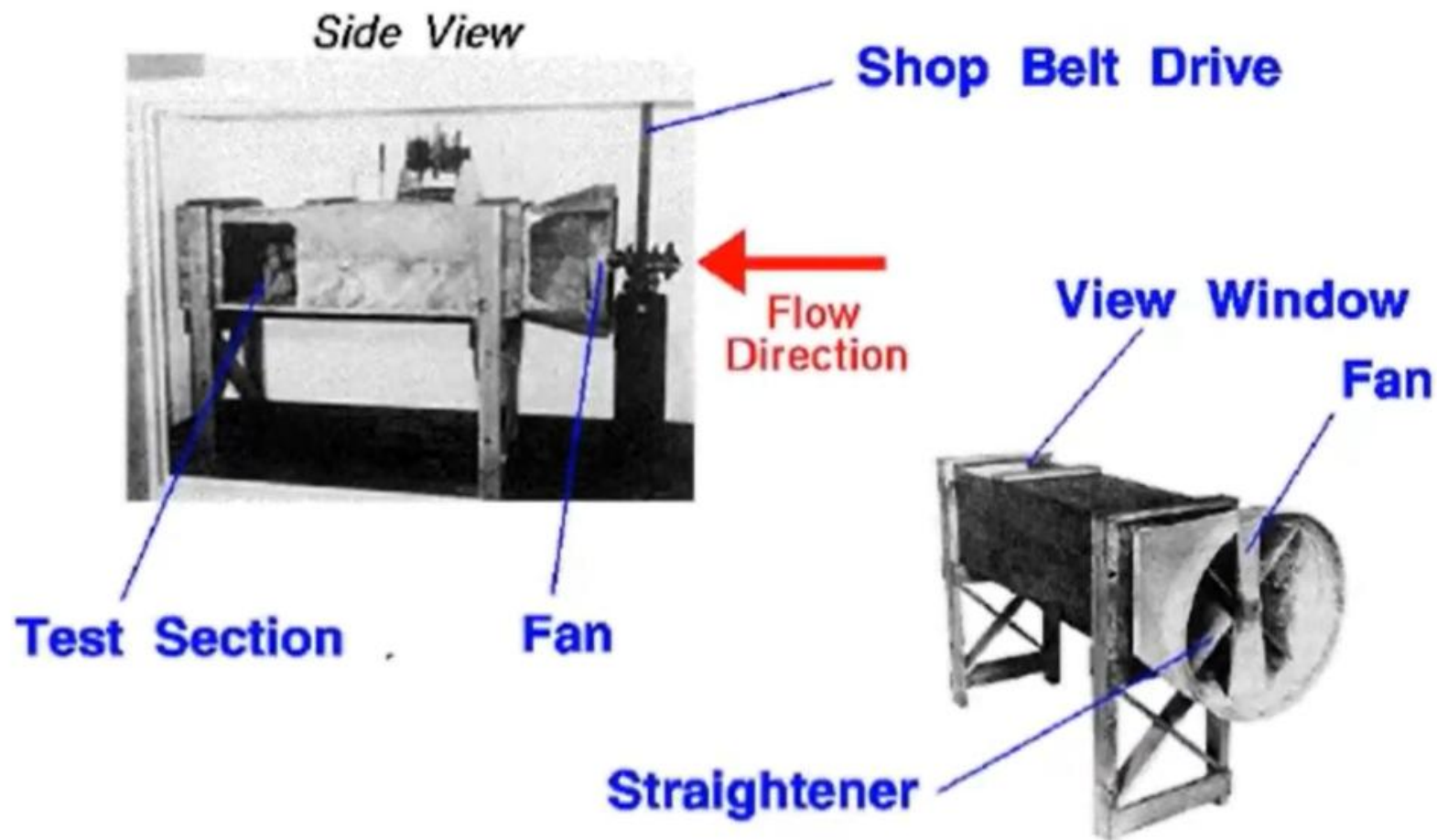


Figure 1.2 Wind tunnel designed, built, and used by the Wright brothers in Dayton, Ohio, during 1901-1902.

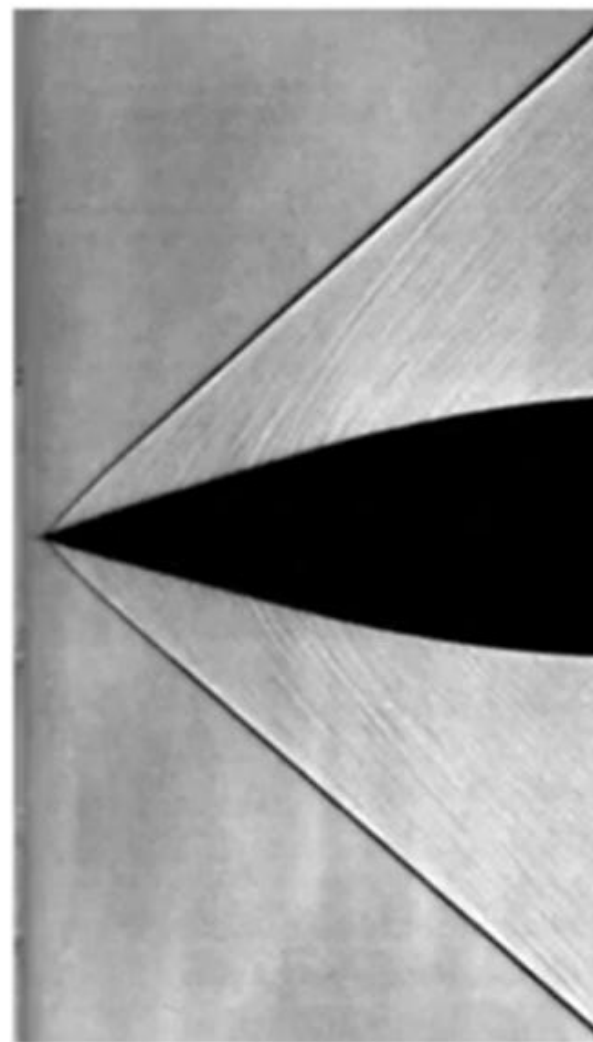
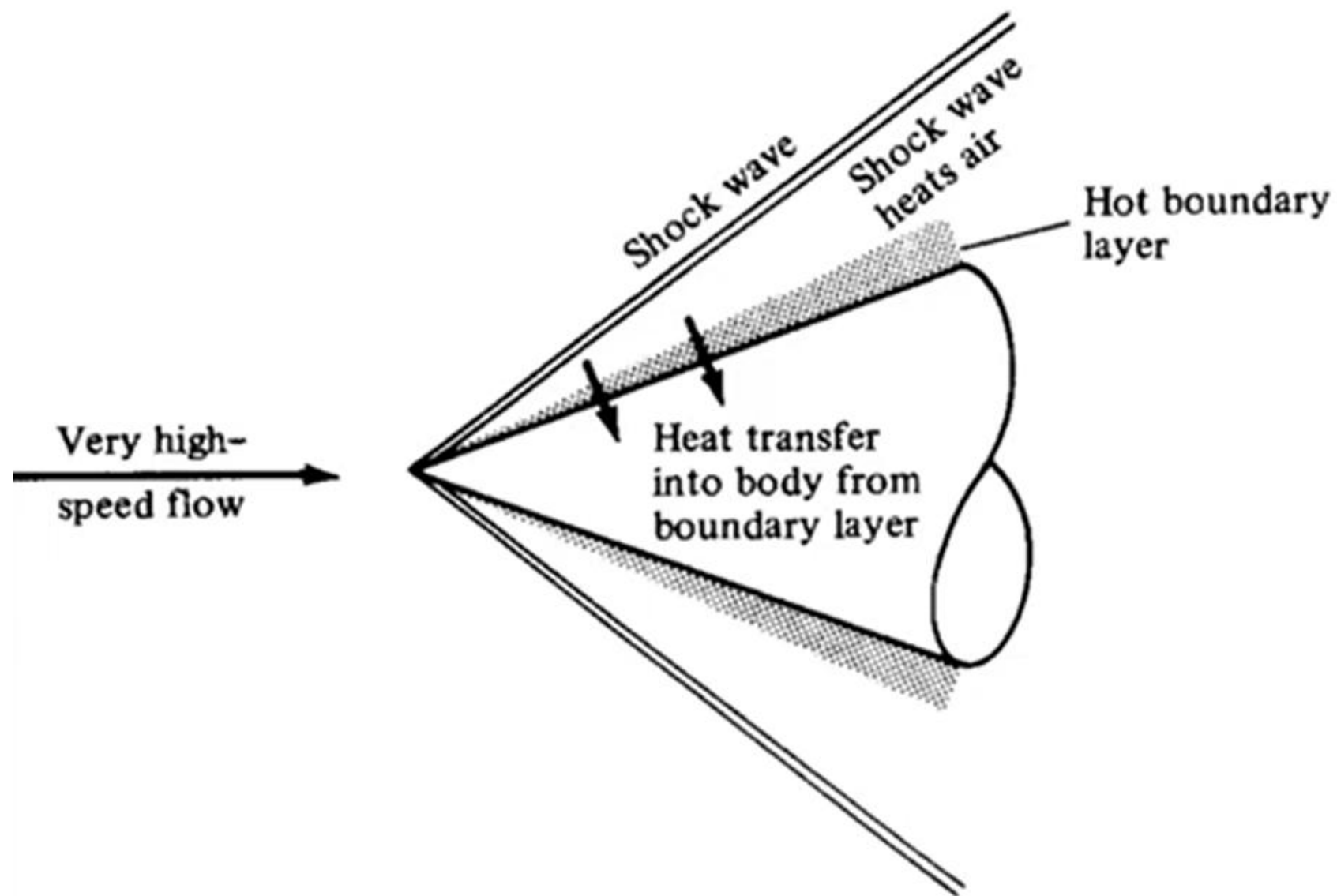
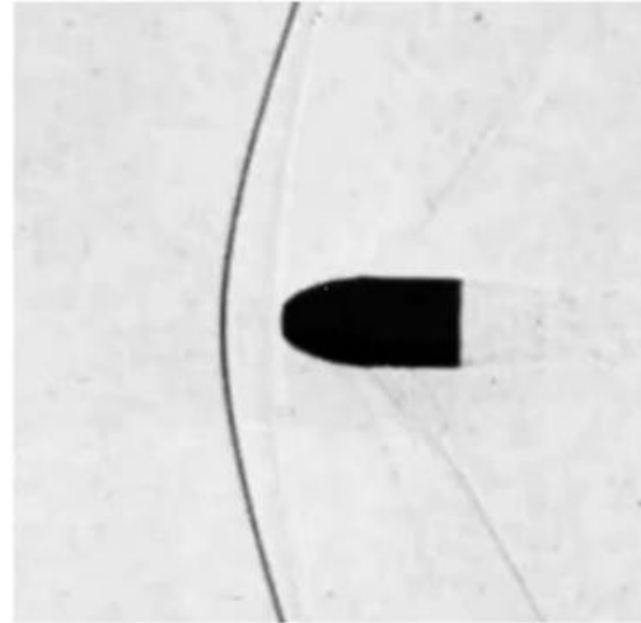
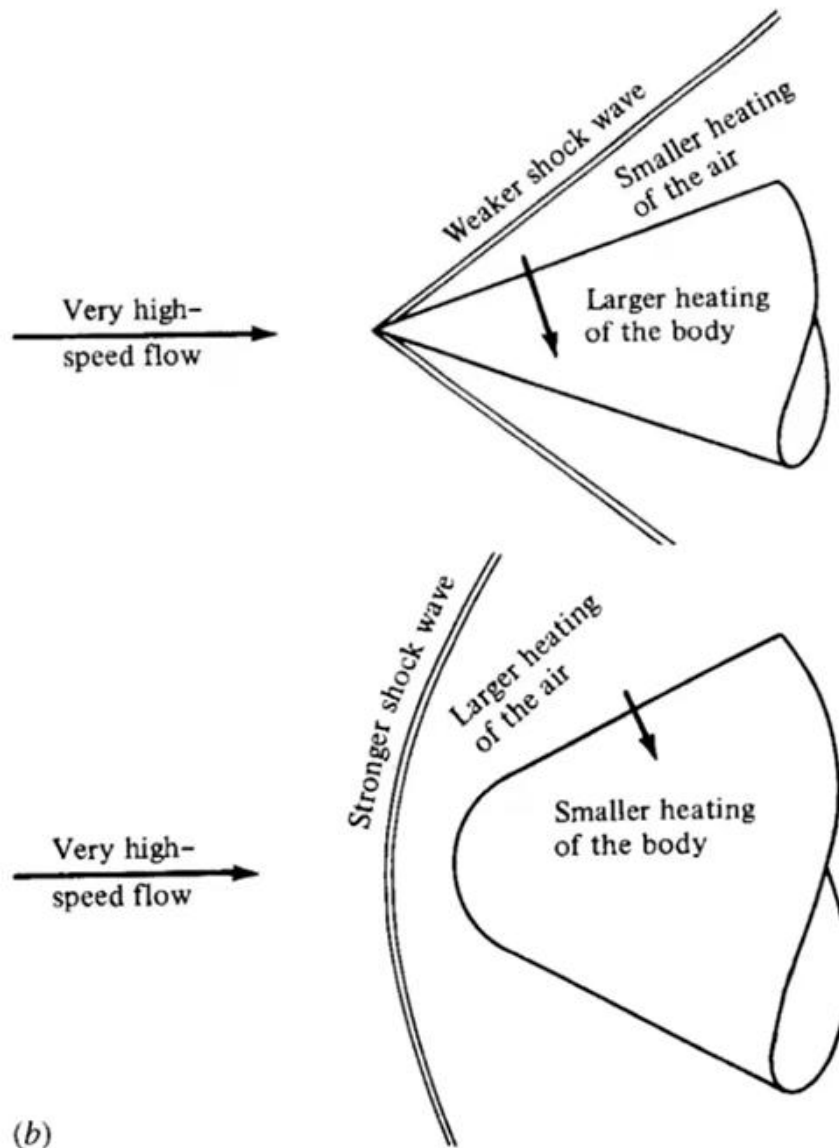
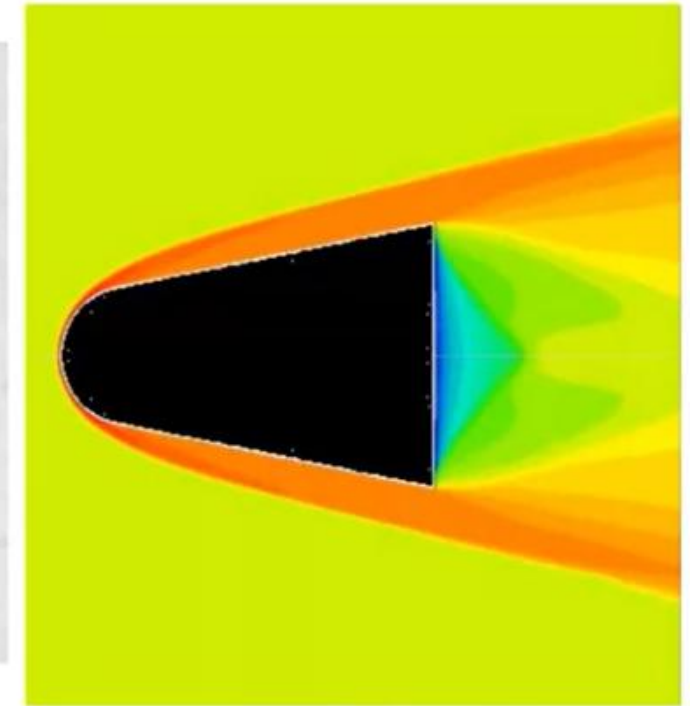


Figure 1.3. Energy of reentry goes into heating both the body and the air around the body.



Experimental

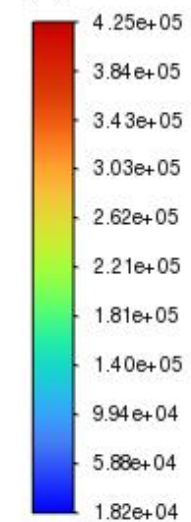


Simulation

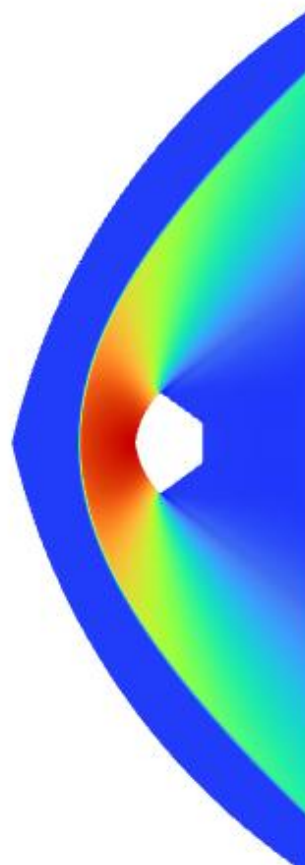
(b)

Figure 1.4 Contrast of aerodynamic heating for slender and blunt reentry vehicles, (a) Slender reentry body, (b) Blunt reentry body.

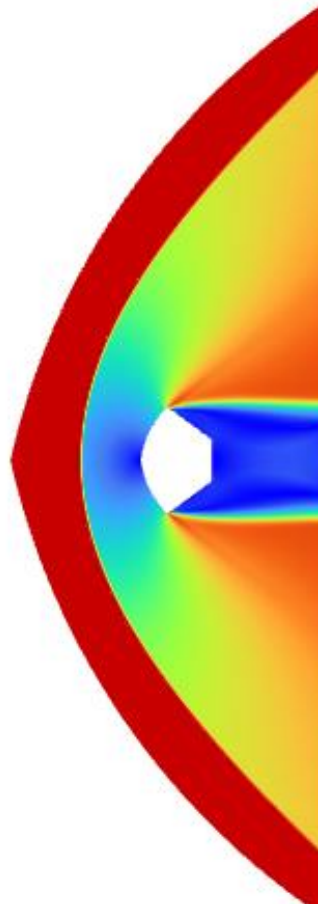
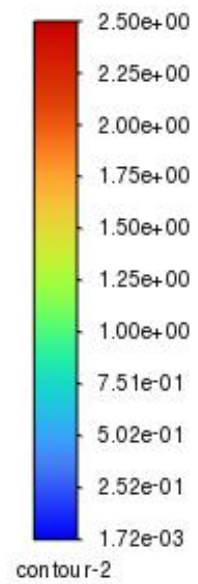
Static Pressure
[Pa]

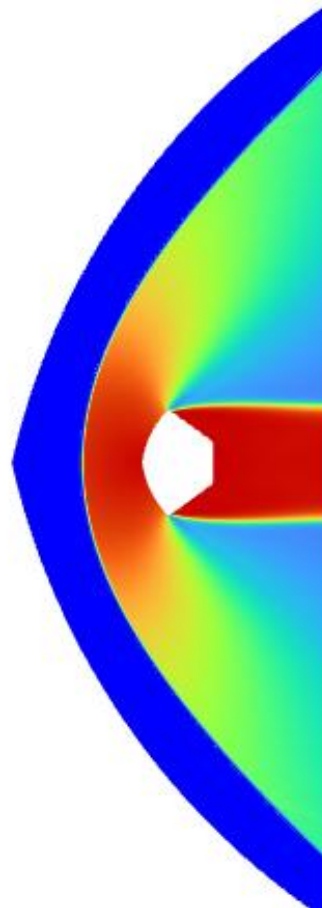
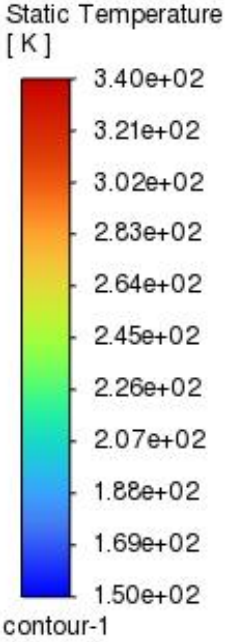


contour-1



Mach Number





Hypersonic Heat Flux Measurement on Double-cone Flow

Mach No. 8.6-8.8. Gas air ($\gamma=1.4$, $R = 287 \text{ J/kg.K}$); $Re_D = 2.08 \times 10^5$, total enthalpy = 0.69 MJ/kg .

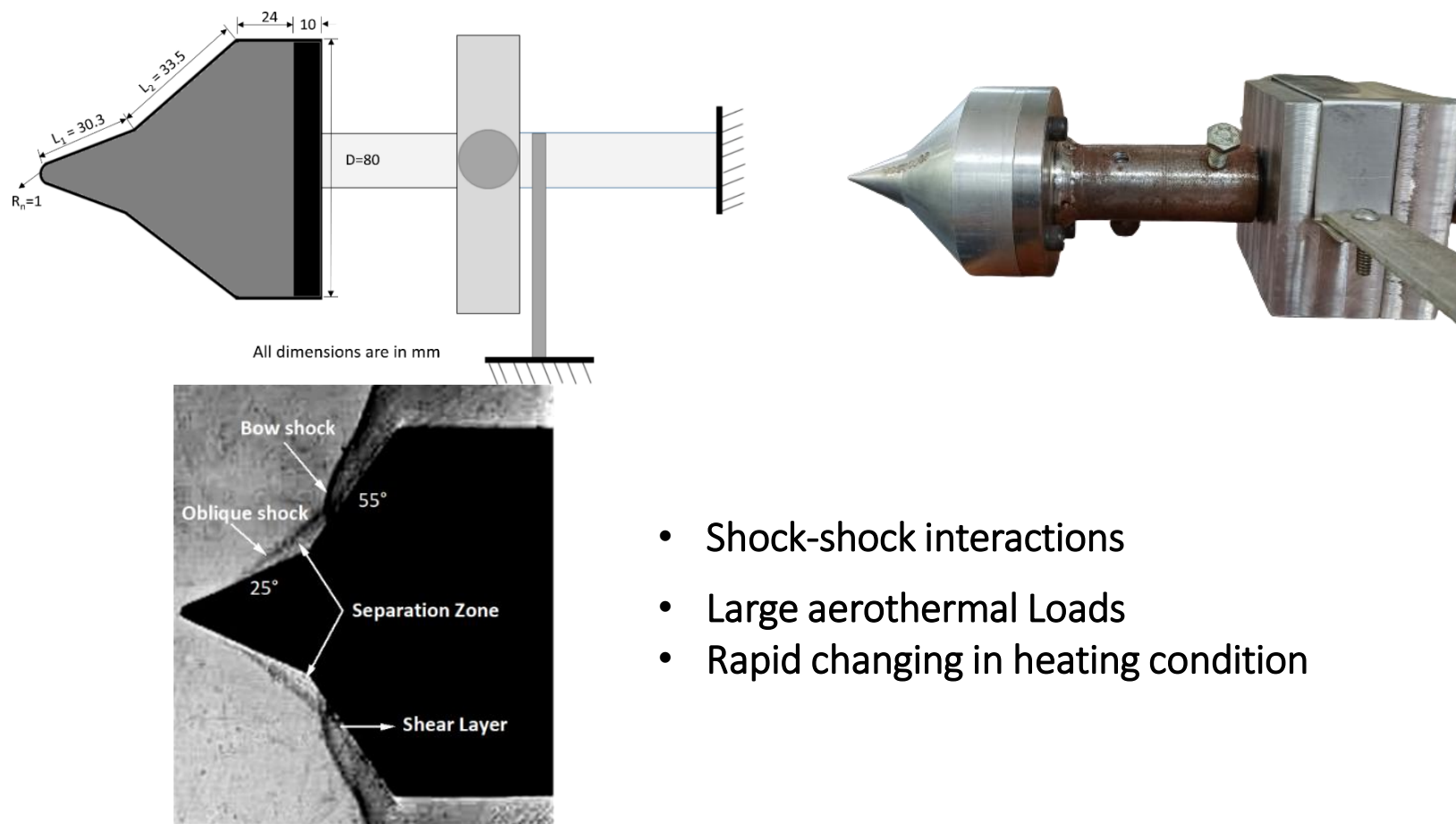
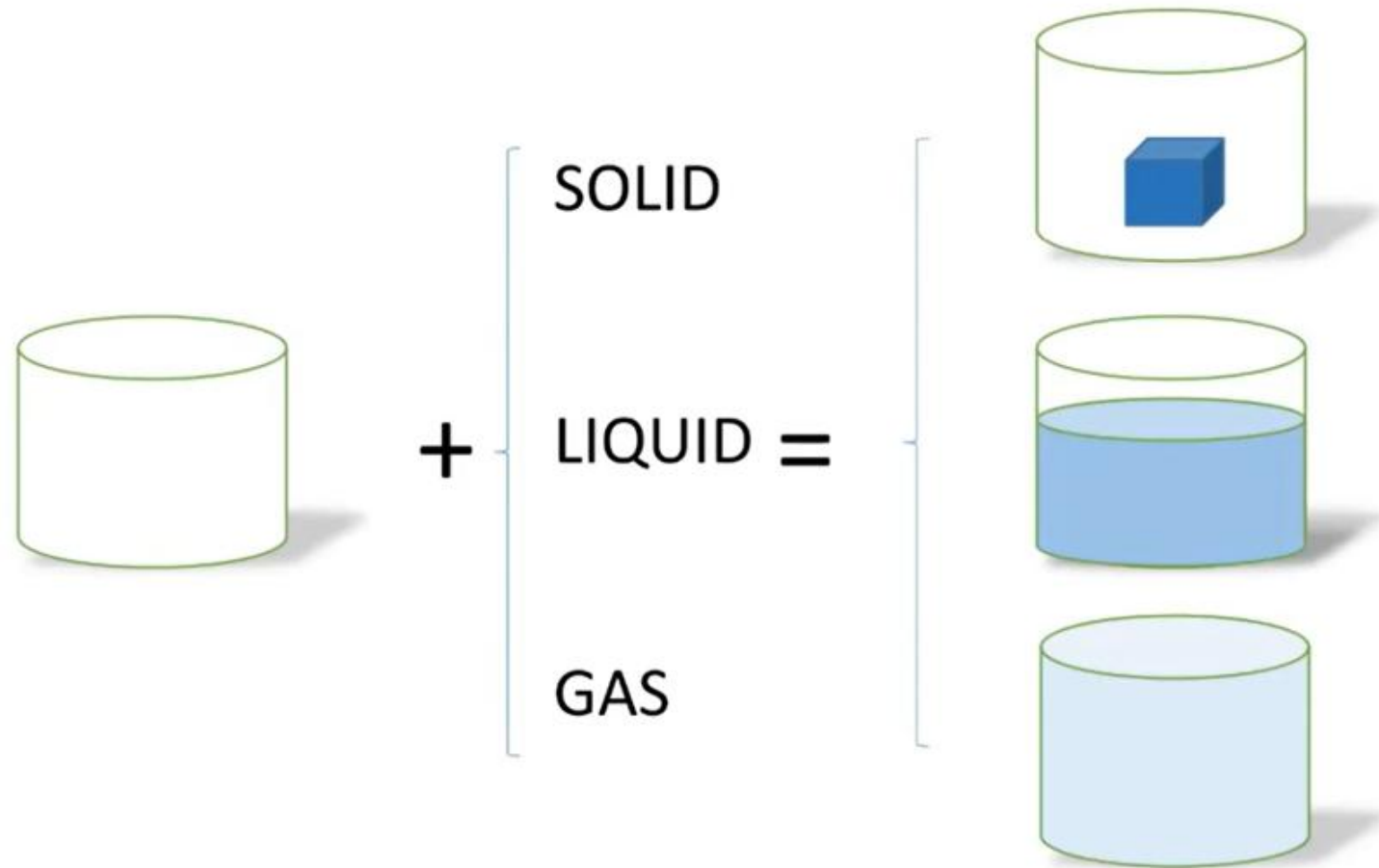


Fig. 8. 25° - 55° double cone model ($L_1/D = 0.38$, $L_2/D = 0.42$, and $R_n/D = 0.0125$); and the flow visualization over a double cone model

Aerodynamics: Classification and Practical Objectives



Divisions of fluid dynamics:

- Hydrodynamics—flow of liquids
- Gas dynamics—flow of gases
- Aerodynamics—flow of air

Some introductory aerodynamic concepts.



Definition of basic aerodynamic quantities; pressure, density, temperature, flow velocity, streamlines



Important to begin an aerodynamic vocabulary

The source of aerodynamic forces and moments, and how to obtain them by integrating surface properties



A starting point for practical applications

Definition of lift, drag, and moment coefficients



Some of the most frequently used terms in aerodynamics

Center of pressure



Where aerodynamic forces act on a body

Dimensional analysis and flow similarity



A powerful instrument to define basic governing parameters; the crux of wind-tunnel testing

Fluid statics



What happens in a fluid when there is no flow

Types of fluid flows: continuum, inviscid/viscous, incompressible/compressible, subsonic, transonic, supersonic, hypersonic



How the science of aerodynamics is subdivided and categorized



The most frequently used words in aerodynamics:

1. pressure

dA = elemental area at B

dF = force on one side of dA due to pressure

Then, the pressure at point B in the fluid is defined as

$$p = \lim \left(\frac{dF}{dA} \right) \quad dA \rightarrow 0$$

2. density

dv = elemental volume around B

dm = mass of fluid inside dv

Then, the density at point B is

$$\rho = \lim_{dv \rightarrow 0} \frac{dm}{dv}$$

3. temperature

$$KE = \frac{3}{2} k T$$

$$P = \rho \cdot R \cdot T$$

4. flow velocity

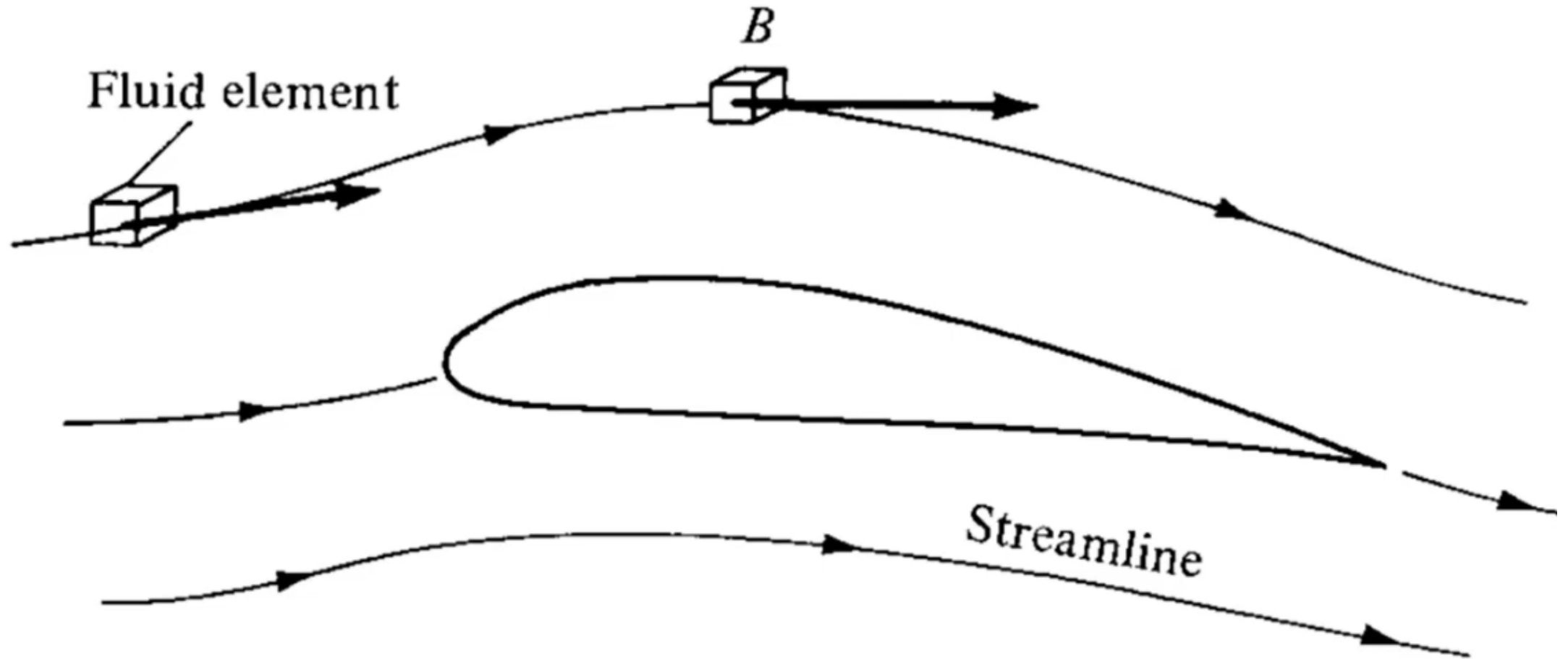
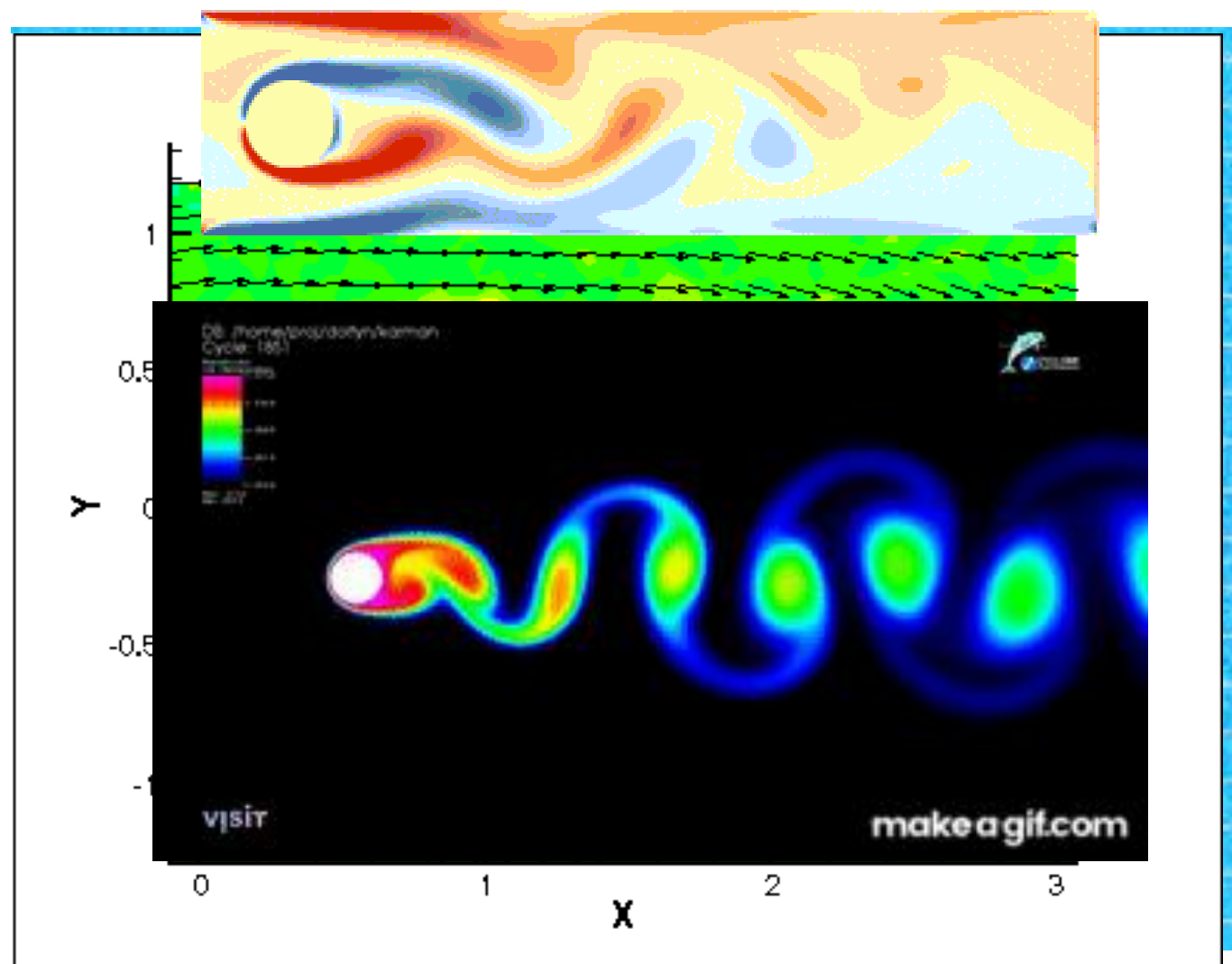
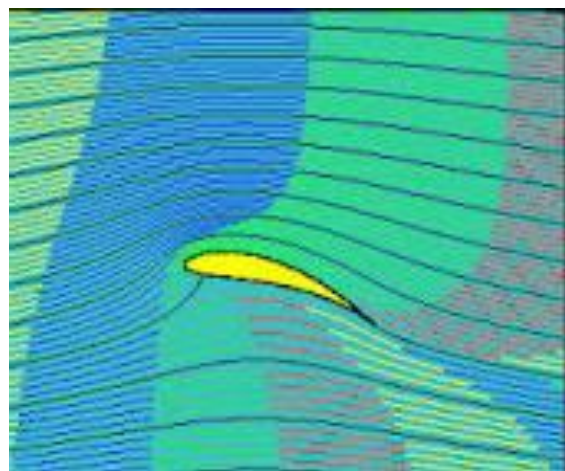
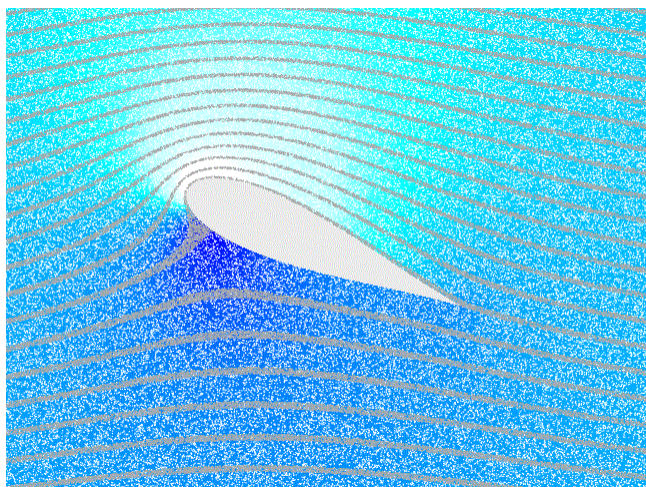
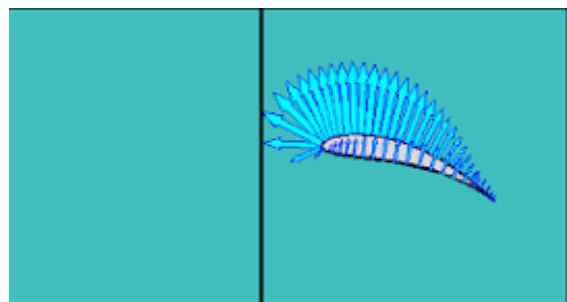
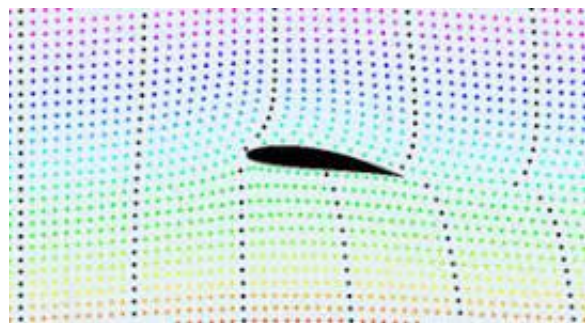
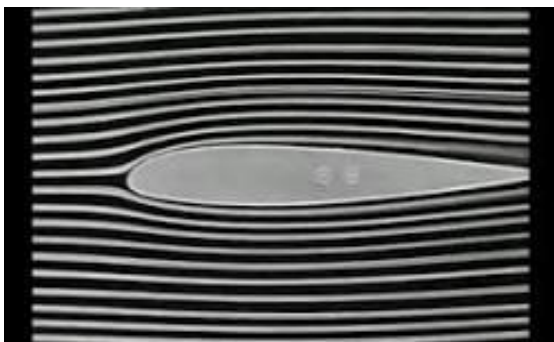


Illustration of flow velocity and streamlines





$$\tau = \lim \left(\frac{dF_f}{dA} \right)$$

$$\tau = \mu \frac{dV}{dy}$$

$$dA \rightarrow 0$$

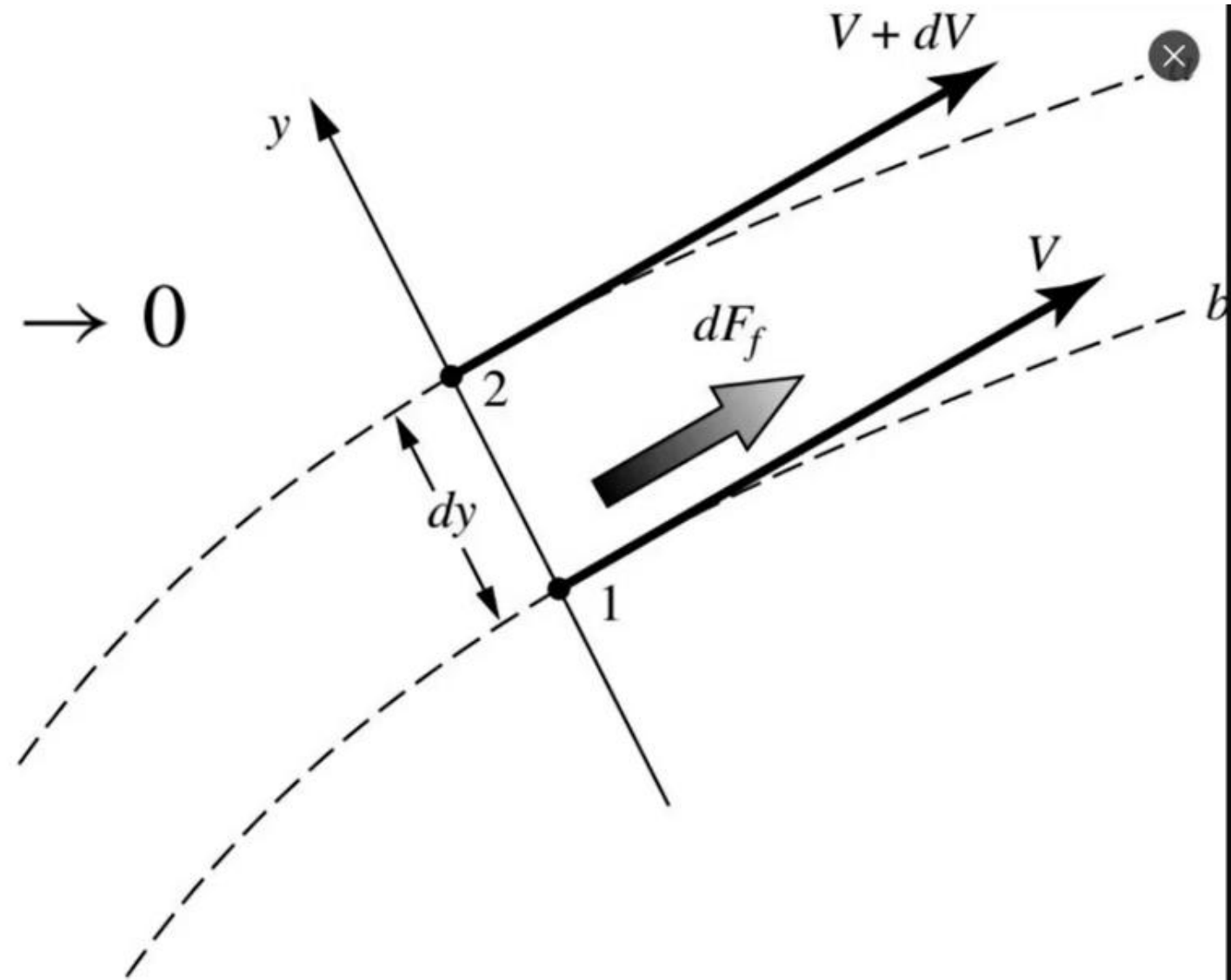
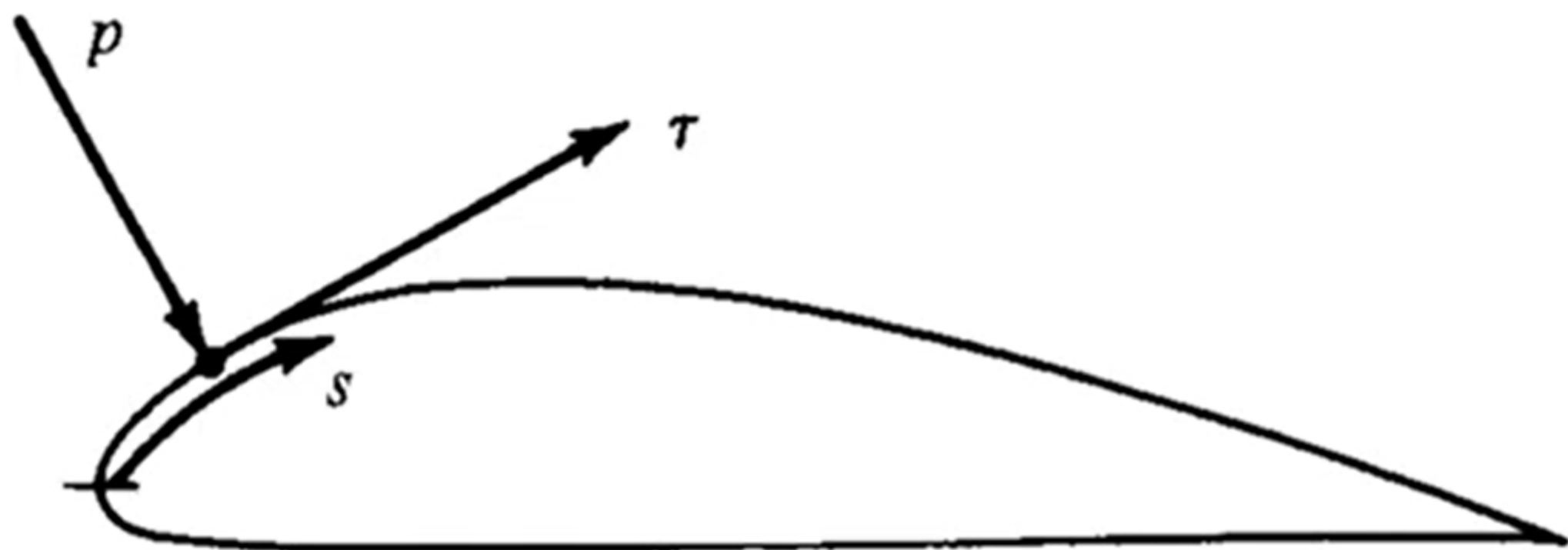


Figure 1.14 Generation of frictional force due to a velocity gradient in a flow.



$p = p(S)$ - surface pressure distribution
 $\tau = \tau(S)$ - surface shear stress distribution

Figure 1.8. Illustration of pressure and shear stress on an aerodynamic surface.

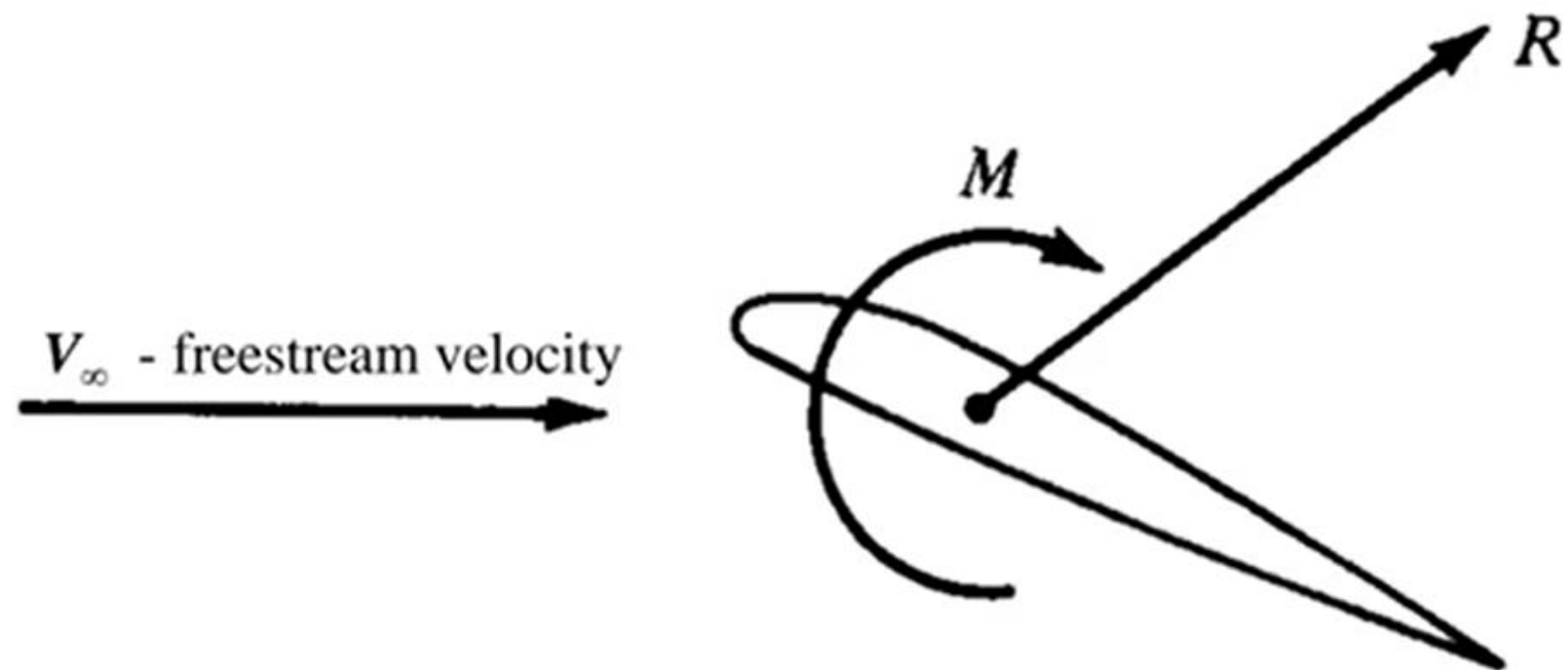


Figure 1.9
Resultant aerodynamic force and moment on the body.

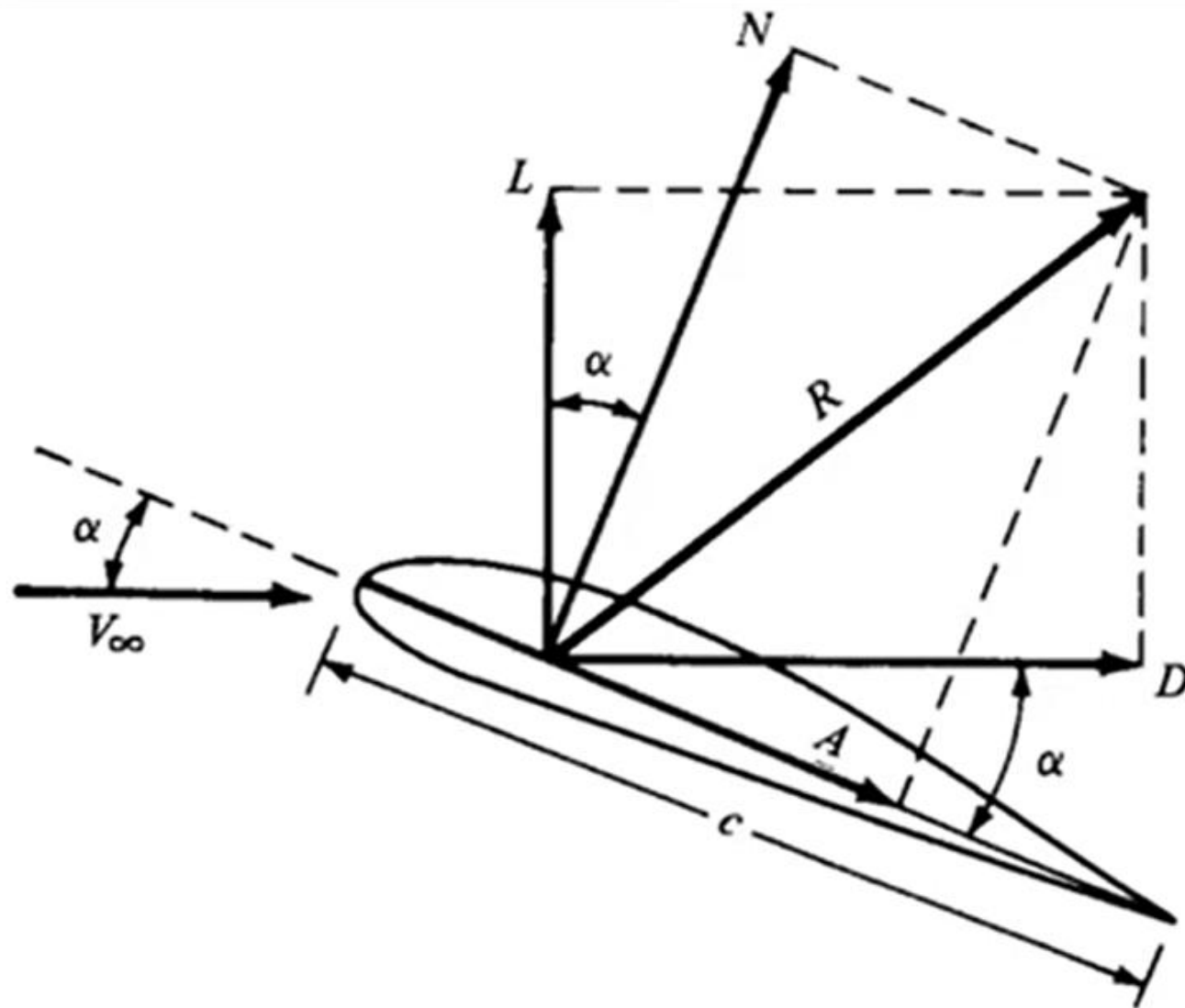


Figure 1.10. Resultant aerodynamic force and the components into which it splits.

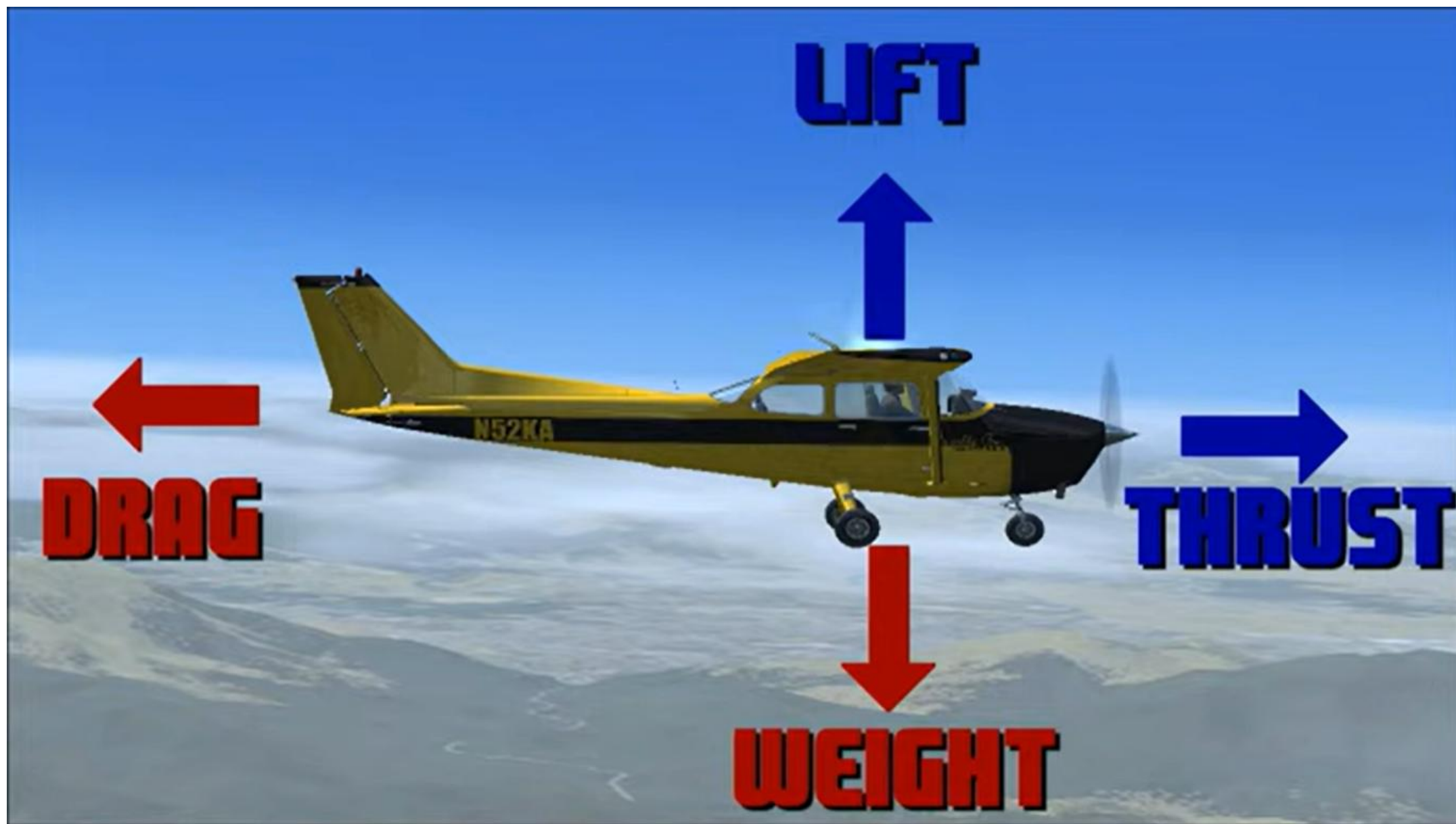
$L \equiv$ lift $\equiv$ component of $\mathbf{R}$
perpendicular to V_∞

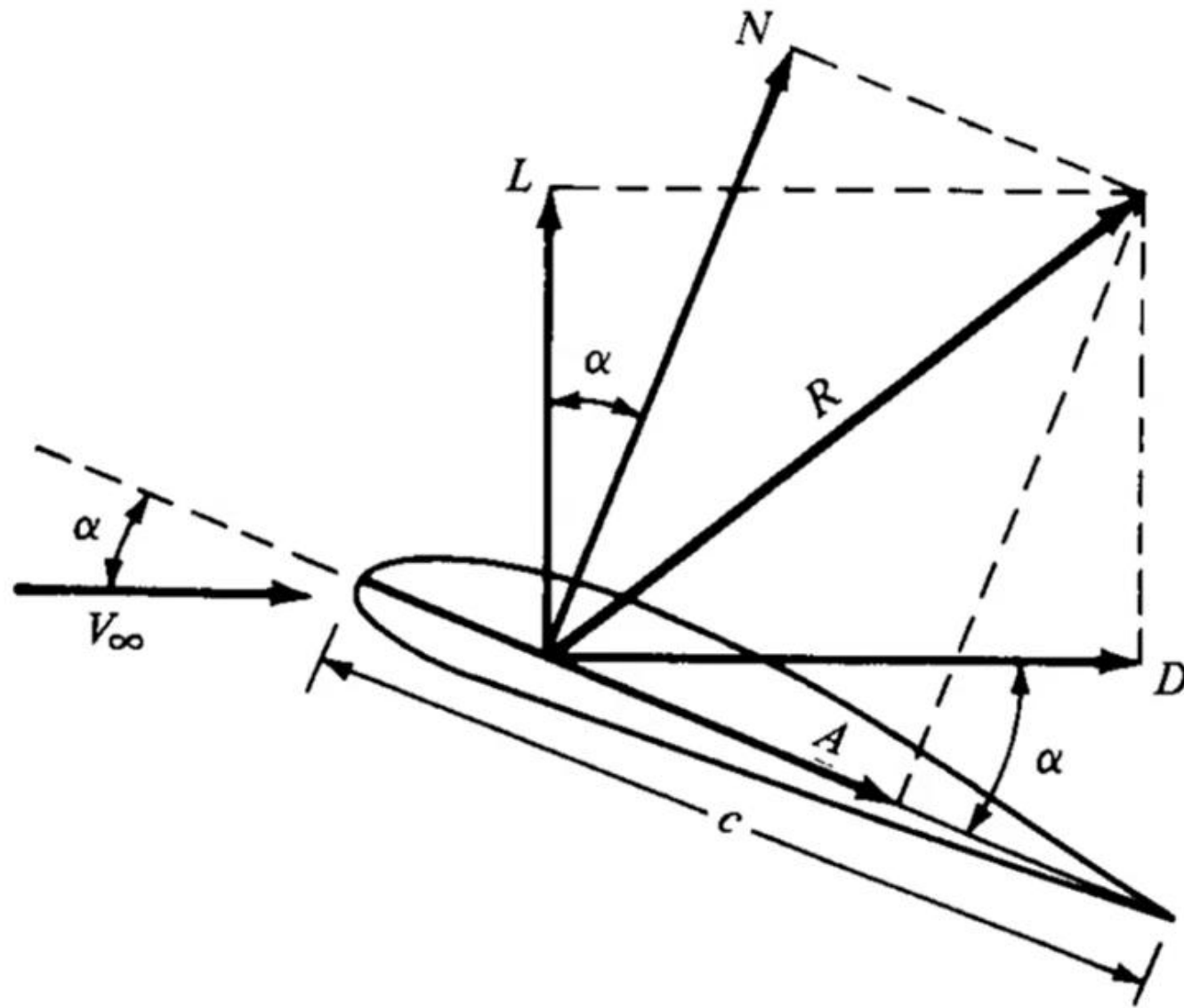
$D \equiv$ drag $\equiv$ component of $\mathbf{R}$ parallel
to V_∞

$\mathbf{N} \equiv$ normal force $\equiv$ component of $\mathbf{R}$
perpendicular to $\mathbf{c}$

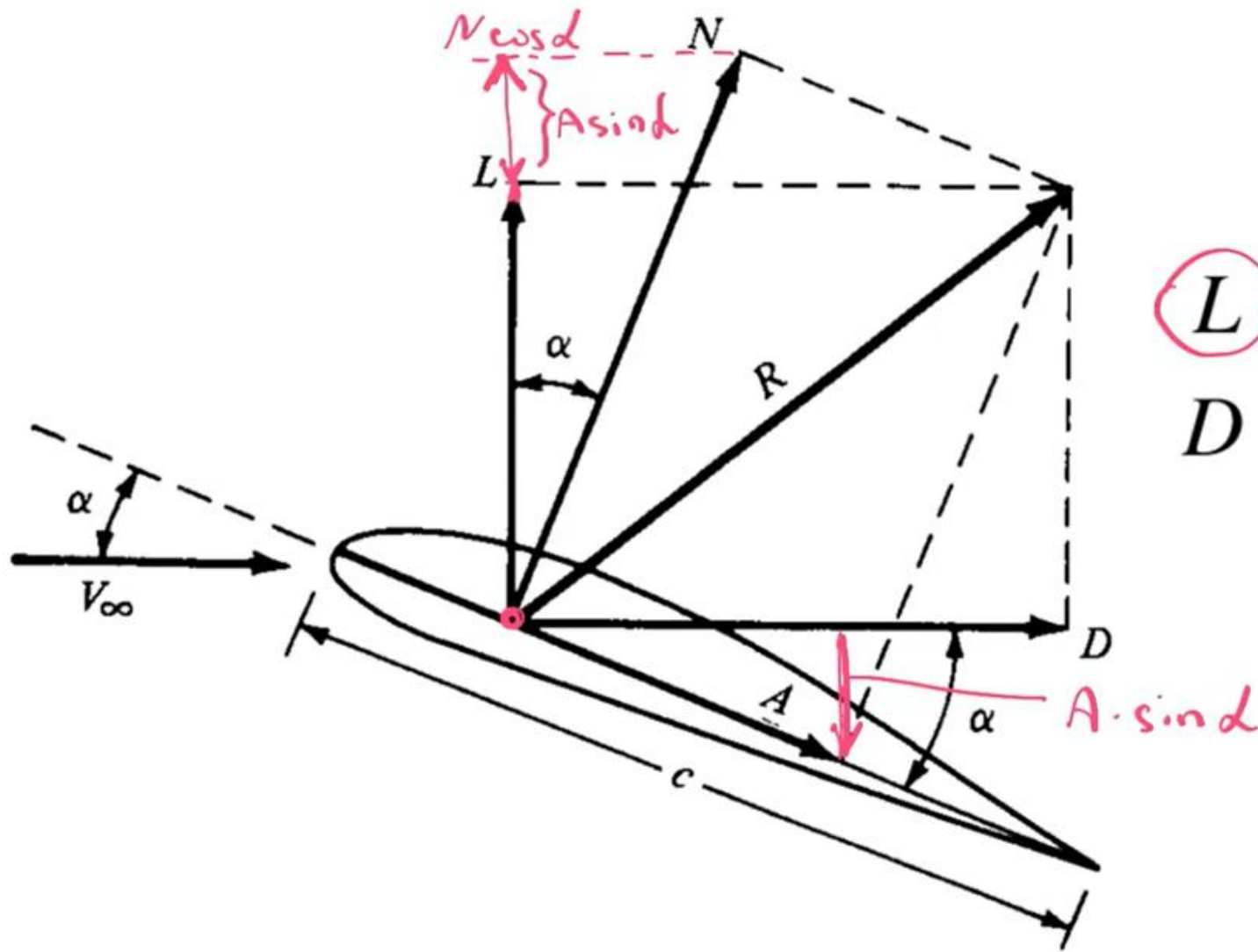
$\mathbf{A} \equiv$ axial force $\equiv$ component of $\mathbf{R}$
parallel to $\mathbf{c}$

$$\bar{\mathbf{R}} = \bar{\mathbf{L}} + \bar{\mathbf{D}} = \bar{\mathbf{N}} + \bar{\mathbf{A}}$$



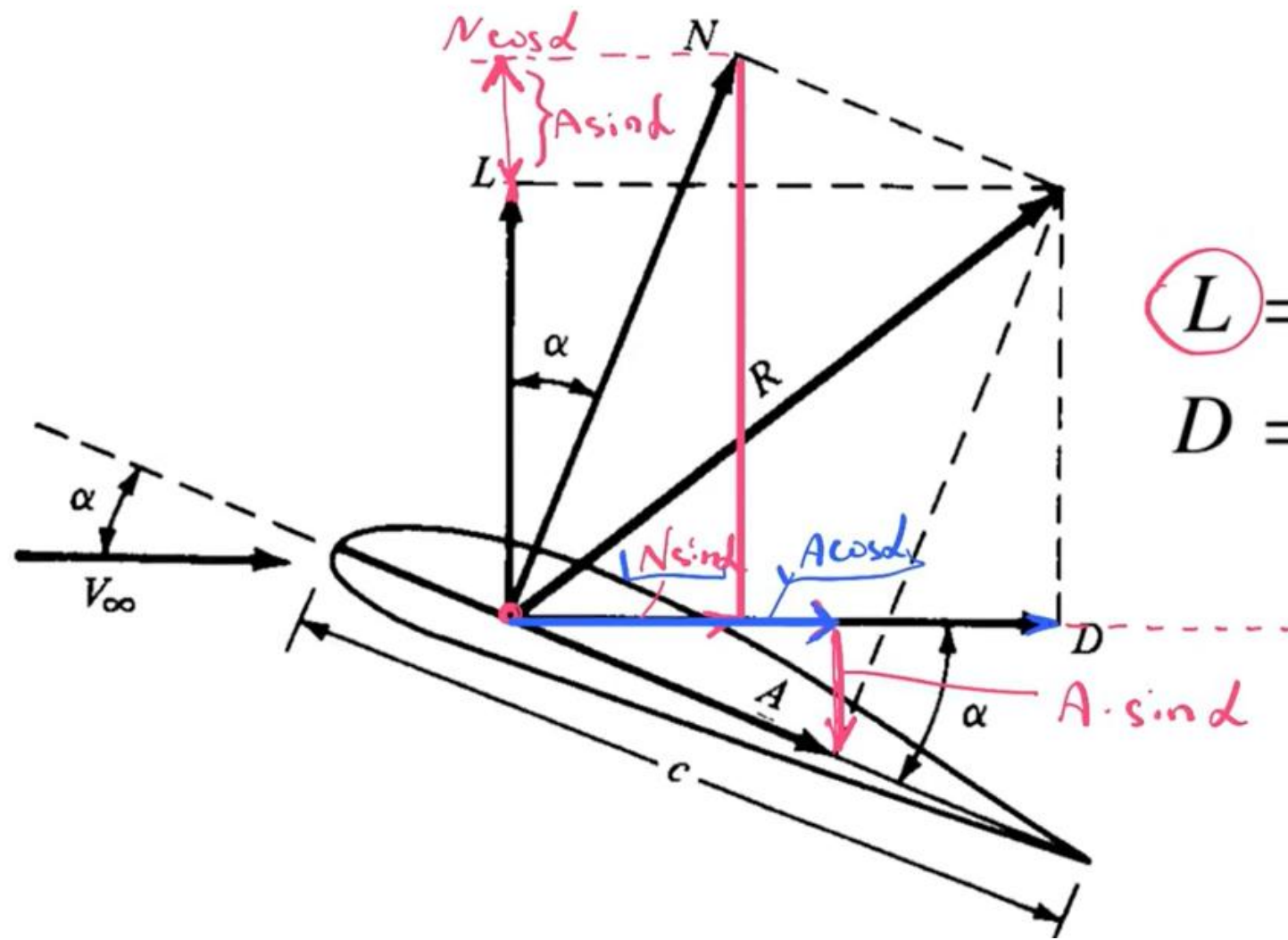


$$L = N \cos \alpha - A \sin \alpha$$
$$D = N \sin \alpha + A \cos \alpha$$



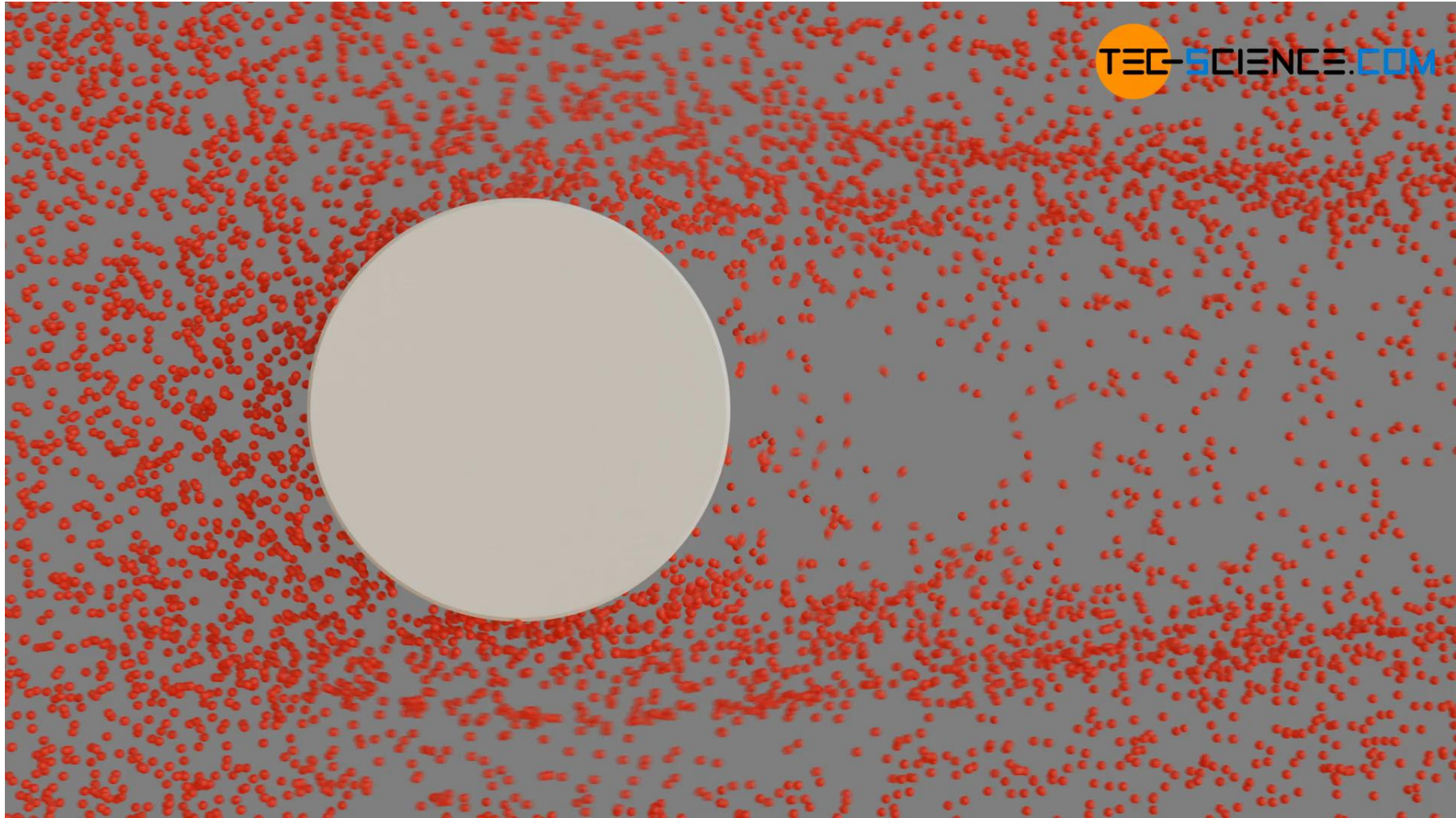
$$\textcircled{L} = \underline{N \cos \alpha} - A \sin \alpha$$

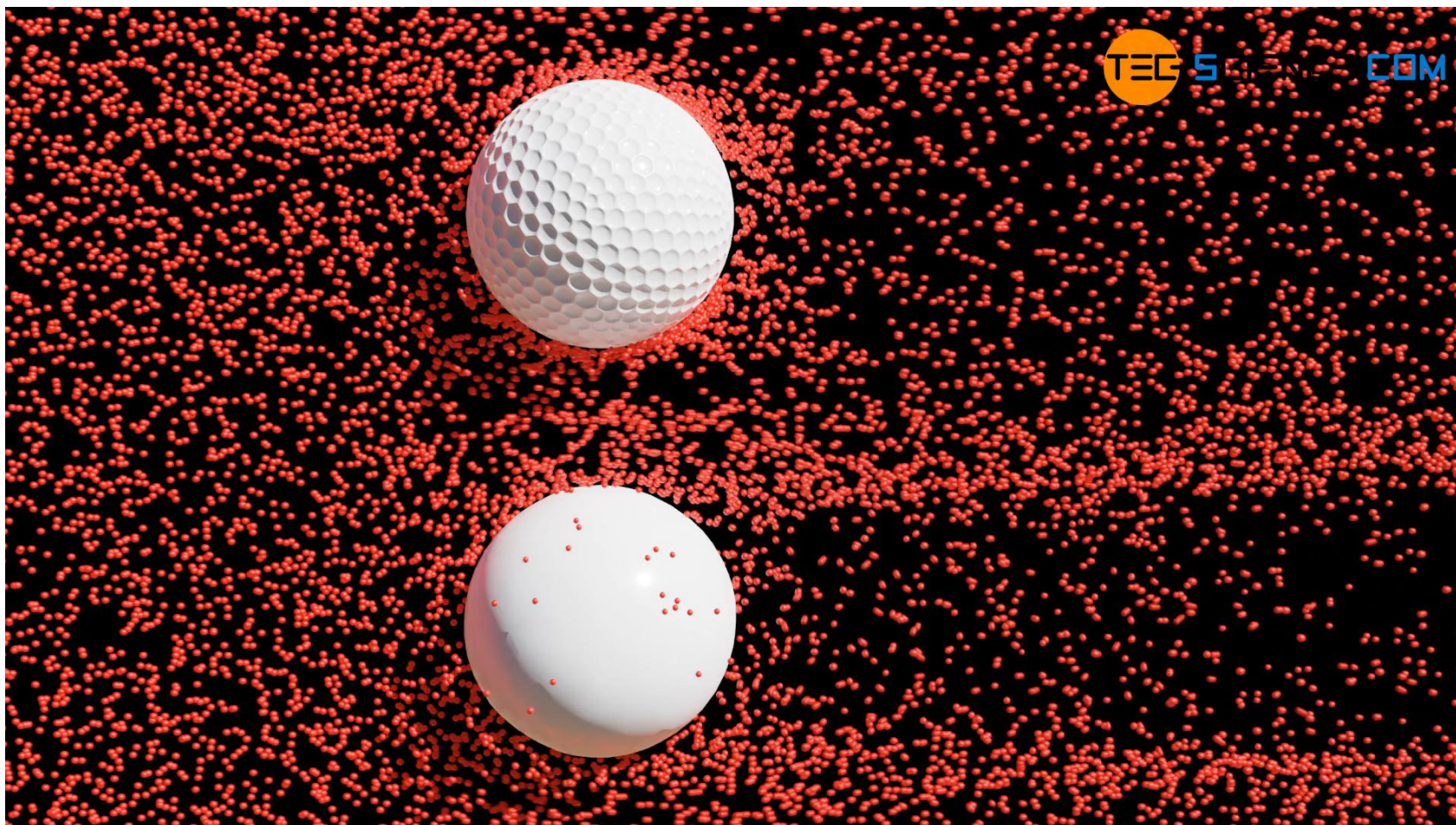
$$D = N \sin \alpha + A \cos \alpha$$



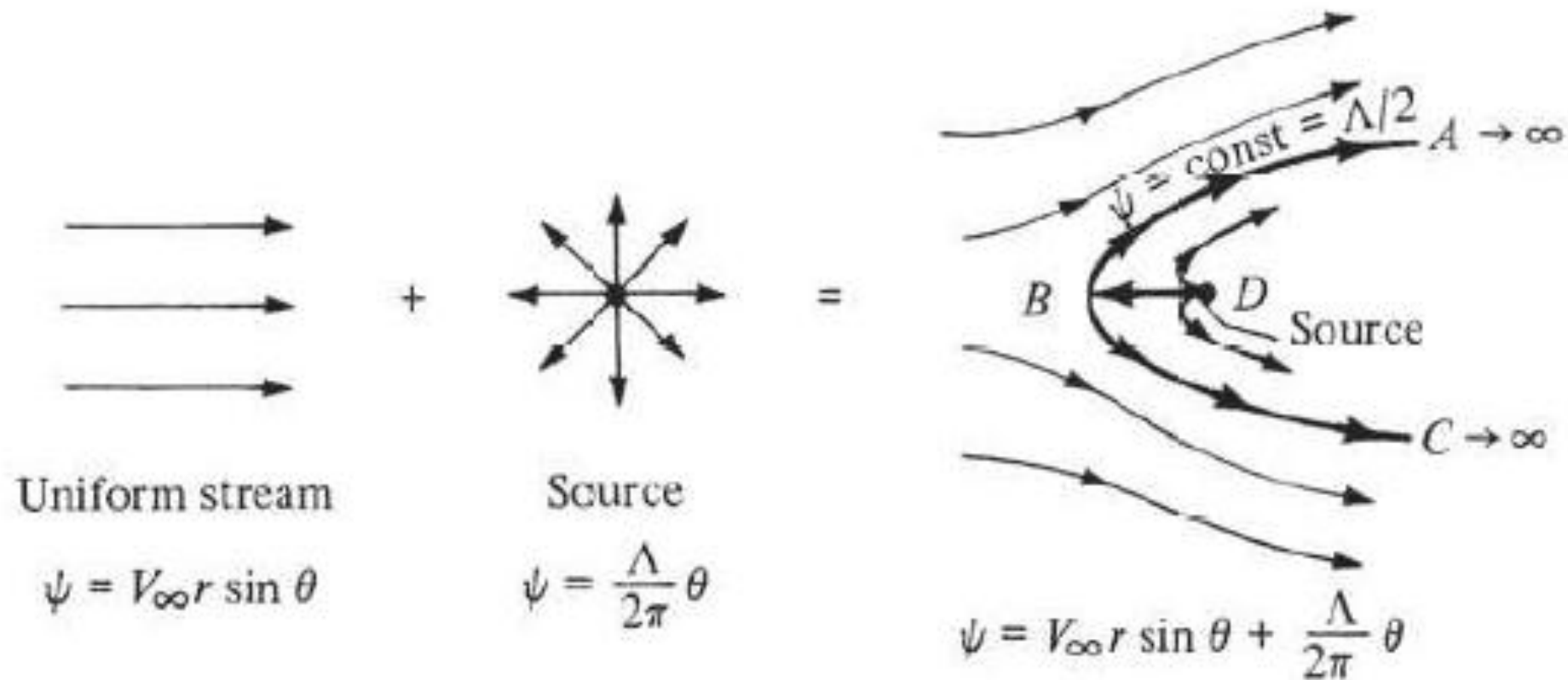
$$\textcircled{L} = \underbrace{N \cos \alpha} - A \sin \alpha$$

$$D = N \sin \alpha + A \cos \alpha$$





COMBINATION OF A UNIFORM FLOW WITH A SOURCE AND SINK



$$\psi = V_{\infty} r \sin \theta + \frac{\Lambda}{2\pi} \theta$$

satisfies Laplace's equation; that describes a viable irrotational, incompressible flow.

The streamlines of the combined flow are obtained from Equation as

$$\psi = V_{\infty} r \sin \theta + \frac{\Lambda}{2\pi} \theta = \text{const}$$

The velocity field is obtained by differentiating Equation

$$V_r = \frac{1}{r} \frac{\partial \psi}{\partial \theta} = V_{\infty} \cos \theta + \frac{\Lambda}{2\pi r}$$
$$V_{\theta} = -\frac{\partial \psi}{\partial r} = -V_{\infty} \sin \theta$$

The stagnation points in the flow can be obtained by setting Equations equal to zero

$$V_{\infty} \cos \theta + \frac{\Lambda}{2\pi r} = 0$$

$$V_{\infty} \sin \theta = 0$$

Solving for r and θ , we find that one stagnation point exists, located at $(r, \theta) = (\Lambda/2\pi V_{\infty}, \pi)$, which is labeled as point B . *That is, the stagnation point is a distance $(\Lambda/2\pi V_{\infty})$ directly upstream of the source.*

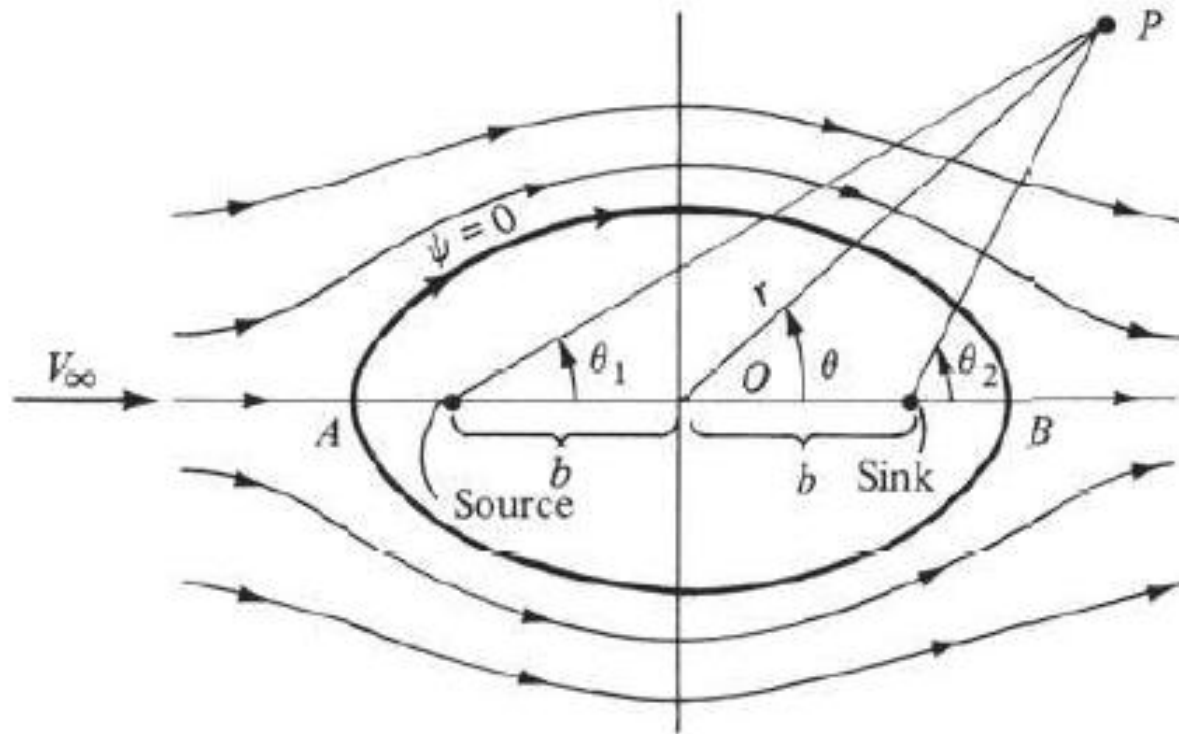
If the coordinates of the stagnation point at B are substituted into Equation

$$\psi = V_{\infty} \frac{\Lambda}{2\pi V_{\infty}} \sin \pi + \frac{\Lambda}{2\pi} \pi = \text{const}$$

$$\psi = \frac{\Lambda}{2} = \text{const}$$

Hence, the streamline that goes through the stagnation point is described by $\psi = \Lambda/2$. This streamline is shown as curve ABC .

COMBINATION OF A UNIFORM FLOW WITH A SOURCE AND SINK



The stream function for the combined flow at any point P with coordinates (r, ϑ) is obtained from Equations

$$\psi = V_\infty r \sin \theta + \frac{\Lambda}{2\pi} \theta_1 - \frac{\Lambda}{2\pi} \theta_2$$

$$\psi = V_\infty r \sin \theta + \frac{\Lambda}{2\pi} (\theta_1 - \theta_2)$$

Superposition of a uniform flow and a source-sink pair; flow over a Rankine oval

The velocity field is obtained by differentiating Equation (3.80) according to Equations (2.151*a* and *b*). Note from the geometry of Figure 3.23 that θ_1 and θ_2 in Equation (3.80) are functions of r , θ , and b . In turn, by setting $V = 0$, two stagnation points are found, namely, points A and B in Figure 3.23. These stagnation points are located such that (see Problem 3.13)

$$OA = OB = \sqrt{b^2 + \frac{\Lambda b}{\pi V_\infty}}$$

The equation of the streamlines is given by Equation (3.80) as

$$\psi = V_\infty r \sin \theta + \frac{\Lambda}{2\pi}(\theta_1 - \theta_2) = \text{const}$$

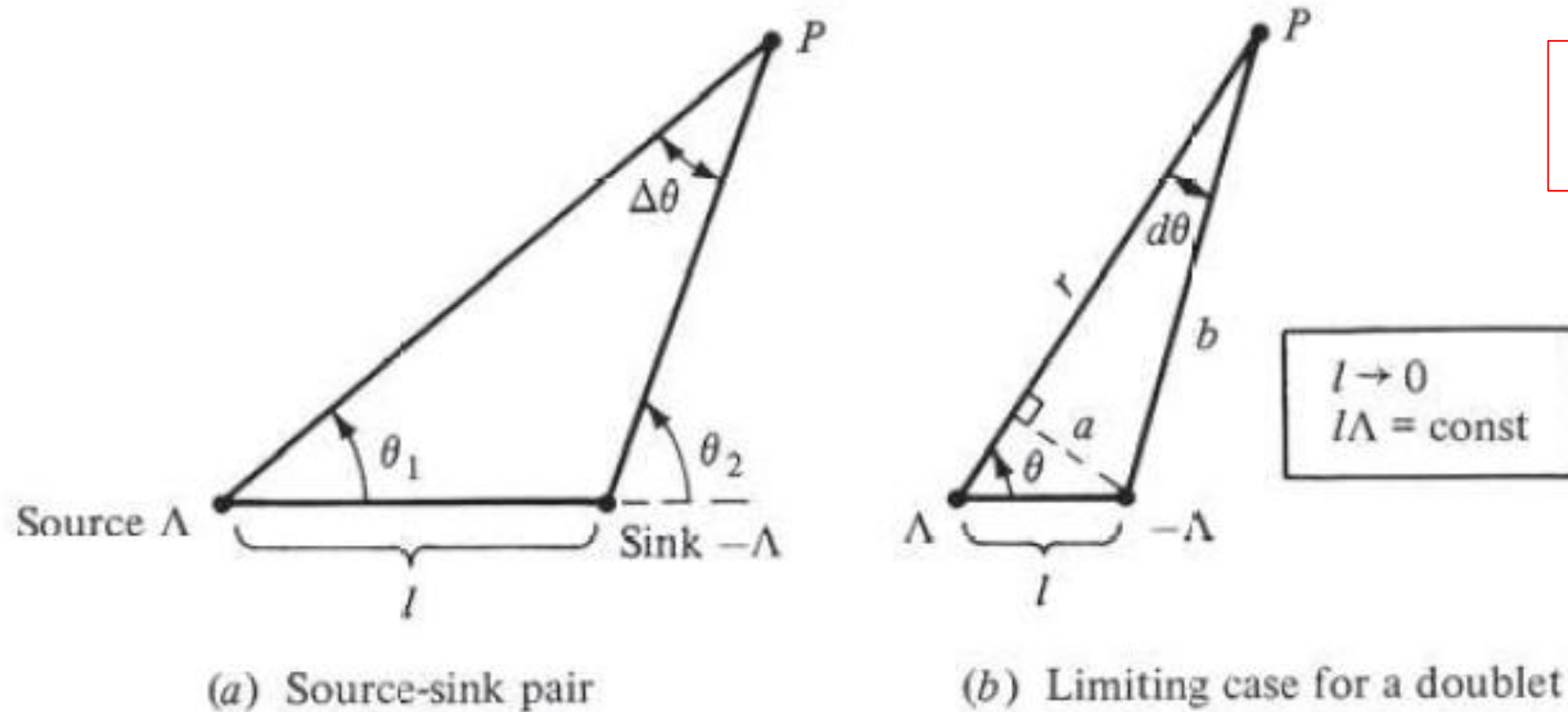
The equation of the specific streamline going through the stagnation points is obtained from Equation (3.82) by noting that $\theta = \theta_1 = \theta_2 = \pi$ at point A and $\theta = \theta_1 = \theta_2 = 0$ at point B . Hence, for the stagnation streamline, Equation (3.82) yields a value of zero for the constant. Thus, the stagnation streamline is given by $\psi = 0$, that is,

$$V_{\infty} r \sin \theta + \frac{\Lambda}{2\pi} (\theta_1 - \theta_2) = 0$$

DOUBLET FLOW: OUR THIRD ELEMENTARY FLOW

There is a special, degenerate case of a source-sink pair that leads to a singularity called a *doublet*. The doublet is frequently used in the theory of incompressible flow; the purpose of this section is to describe its properties.

Consider a source of strength Λ and a sink of equal (but opposite) strength $-\Lambda$ separated by a distance l , as shown in Figure 3.24a. At any point P in the flow the stream function is



$$\psi = \frac{\Lambda}{2\pi}(\theta_1 - \theta_2) = -\frac{\Lambda}{2\pi}\Delta\theta$$

Eq. (3.84)

Figure 3.24 How a source-sink pair approaches a doublet in the limiting case.

where $\Delta\theta = \theta_2 - \theta_1$ as seen from Figure 3.24a. Equation (3.84) is the stream function for a source-sink pair separated by the distance l .

Now in Figure 3.24a, let the distance l approach zero while the absolute magnitudes of the strengths of the source and sink increase in such a fashion that the product $l\Lambda$ remains constant. This limiting process is shown in Figure 3.24b. In the limit, as $l \rightarrow 0$ while $l\Lambda$ remains constant, we obtain a special flow pattern defined as a *doublet*. The *strength* of the doublet is denoted by κ and is defined as $\kappa \equiv l\Lambda$. The stream function for a doublet is obtained from Equation (3.84) as follows:

$$\psi = \lim_{\substack{l \rightarrow 0 \\ \kappa = l\Lambda = \text{const}}} \left(-\frac{\Lambda}{2\pi} d\theta \right) \quad (3.85)$$

where in the limit $\Delta\theta \rightarrow d\theta \rightarrow 0$. (Note that the source strength Λ approaches an infinite value in the limit.) In Figure 3.24*b*, let r and b denote the distances to point P from the source and sink, respectively. Draw a line from the sink perpendicular to r , and denote the length along this line by a . For an infinitesimal $d\theta$, the geometry of Figure 3.24*b* yields

$$a = l \sin \theta$$

$$b = r - l \cos \theta$$

$$d\theta = \frac{a}{b}$$

Hence,

$$d\theta = \frac{a}{b} = \frac{l \sin \theta}{r - l \cos \theta} \quad (3.86)$$

Substituting Equation (3.86) into (3.85), we have

$$\psi = \lim_{\substack{l \rightarrow 0 \\ \kappa = \text{const}}} \left(-\frac{\Lambda}{2\pi} \frac{l \sin \theta}{r - l \cos \theta} \right)$$

Substituting Equation (3.86) into (3.85), we have

$$\psi = \lim_{\substack{l \rightarrow 0 \\ \kappa = \text{const}}} \left(-\frac{\Lambda}{2\pi} \frac{l \sin \theta}{r - l \cos \theta} \right)$$

or

$$\psi = \lim_{\substack{l \rightarrow 0 \\ \kappa = \text{const}}} \left(-\frac{\kappa}{2\pi} \frac{\sin \theta}{r - l \cos \theta} \right)$$

or

$$\boxed{\psi = -\frac{\kappa}{2\pi} \frac{\sin \theta}{r}} \quad (3.87)$$

Equation (3.87) is the stream function for a doublet. In a similar fashion, the velocity potential for a doublet is given by (see Problem 3.14)

$$\boxed{\phi = \frac{\kappa}{2\pi} \frac{\cos \theta}{r}} \quad (3.88)$$

The streamlines of a doublet flow are obtained from Equation (3.87):

$$\psi = -\frac{\kappa}{2\pi} \frac{\sin \theta}{r} = \text{const} = c$$

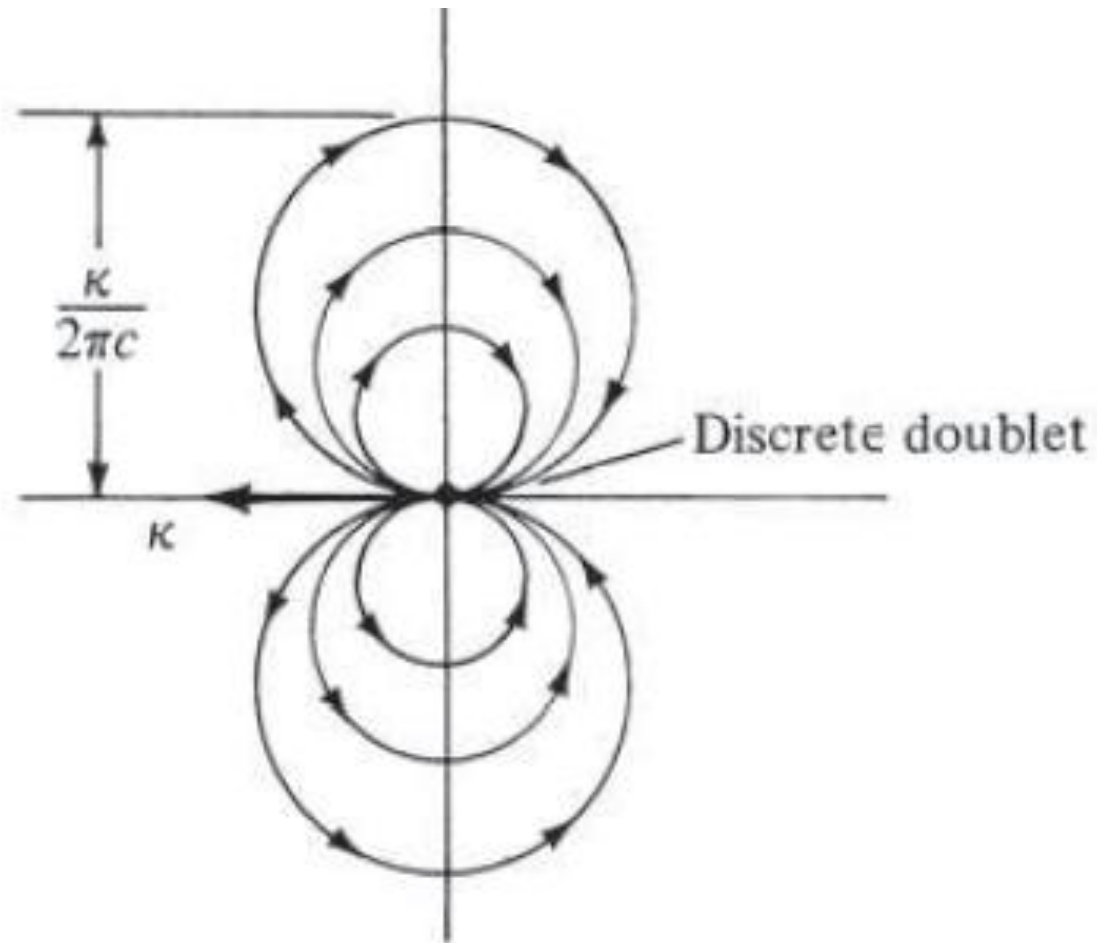
or

$$r = -\frac{\kappa}{2\pi c} \sin \theta \quad (3.89)$$

Equation (3.89) gives the equation for the streamlines in doublet flow. Recall from analytic geometry that the following equation in polar coordinates

$$r = d \sin \theta \quad (3.90)$$

is a circle with a diameter d on the vertical axis and with the center located $d/2$ directly above the origin. Comparing Equations (3.89) and (3.90), we see that the streamlines for a doublet are a family of circles with diameter $\kappa/2\pi c$, as sketched in Figure 3.25. The different circles correspond to different values of the parameter c . Note that in Figure 3.24 we placed the source to the left of the sink; hence, in Figure 3.25 the direction of flow is out of the origin to



$$\psi = -\frac{\kappa}{2\pi} \frac{\sin \theta}{r}$$

$$\phi = \frac{\kappa}{2\pi} \frac{\cos \theta}{r}$$

$$\psi = -\frac{\kappa}{2\pi} \frac{\sin \theta}{r} = \text{const} = c$$

$$r = -\frac{\kappa}{2\pi c} \sin \theta$$

$$r = d \sin \theta$$

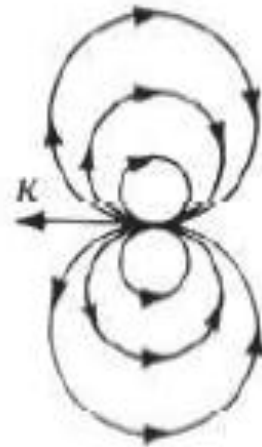
NONLIFTING FLOW OVER A CIRCULAR CYLINDER



Uniform flow

$$\psi = V_{\infty} r \sin \theta$$

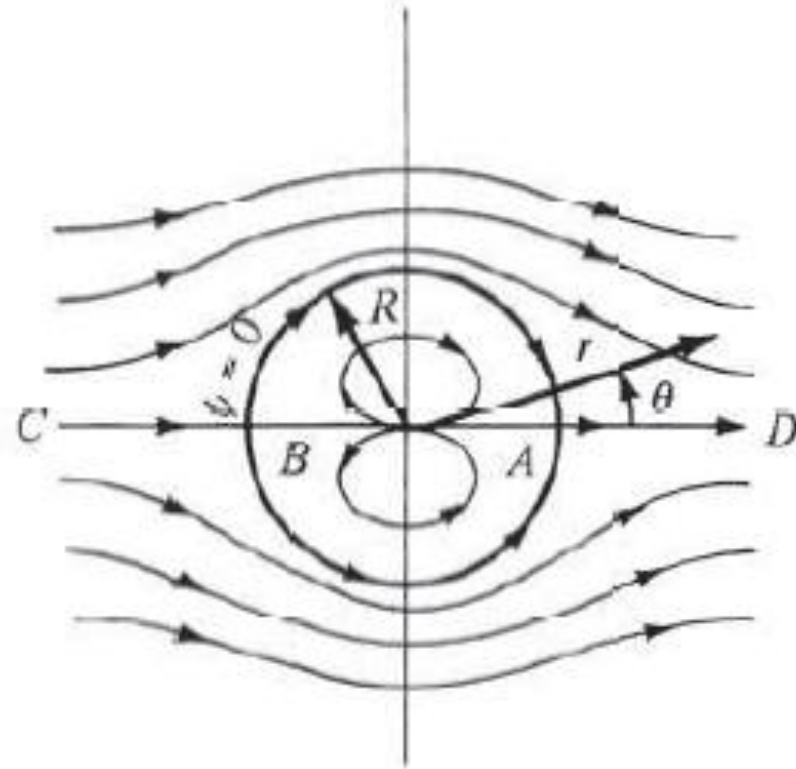
+



Doublet

$$\psi = \frac{-\kappa}{2\pi} \frac{\sin \theta}{r}$$

=



Flow over a cylinder

$$\psi = V_{\infty} r \sin \theta - \frac{\kappa}{2\pi} \frac{\sin \theta}{r}$$

$$\psi = V_{\infty} r \sin \theta - \frac{\kappa}{2\pi} \frac{\sin \theta}{r}$$

or

$$\psi = V_{\infty} r \sin \theta \left(1 - \frac{\kappa}{2\pi V_{\infty} r^2} \right)$$

Let $R^2 \equiv \kappa/2\pi V_{\infty}$. Then Equation (3.91) can be written as

$$\psi = (V_{\infty} r \sin \theta) \left(1 - \frac{R^2}{r^2} \right)$$

$$V_r = \frac{1}{r} \frac{\partial \psi}{\partial \theta} = \frac{1}{r} (V_{\infty} r \cos \theta) \left(1 - \frac{R^2}{r^2} \right)$$

$$V_r = \left(1 - \frac{R^2}{r^2} \right) V_{\infty} \cos \theta$$

$$V_{\theta} = -\frac{\partial \psi}{\partial r} = - \left[(V_{\infty} r \sin \theta) \frac{2R^2}{r^3} + \left(1 - \frac{R^2}{r^2} \right) (V_{\infty} \sin \theta) \right]$$

$$V_{\theta} = - \left(1 + \frac{R^2}{r^2} \right) V_{\infty} \sin \theta$$

$$\left(1 - \frac{R^2}{r^2} \right) V_{\infty} \cos \theta = 0$$

$$\left(1 + \frac{R^2}{r^2} \right) V_{\infty} \sin \theta = 0$$

$$\psi = (V_{\infty} r \sin \theta) \left(1 - \frac{R^2}{r^2} \right) = 0$$

$$R = \sqrt{\frac{\kappa}{2\pi V_{\infty}}}$$

$$V_r = 0$$

$$V_{\theta} = -2V_{\infty} \sin \theta$$

The pressure coefficient is given by Equation

$$C_p = 1 - \left(\frac{V}{V_\infty} \right)^2$$

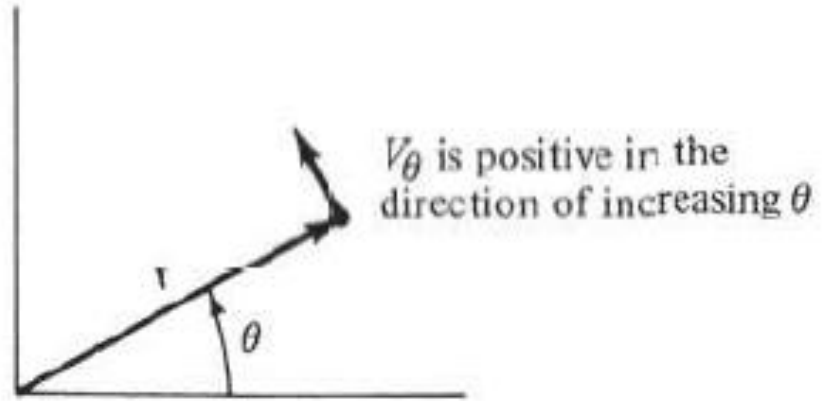
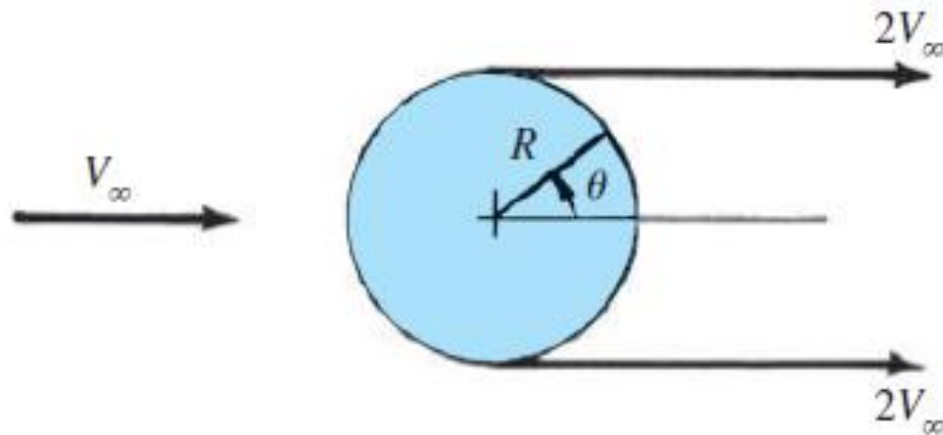
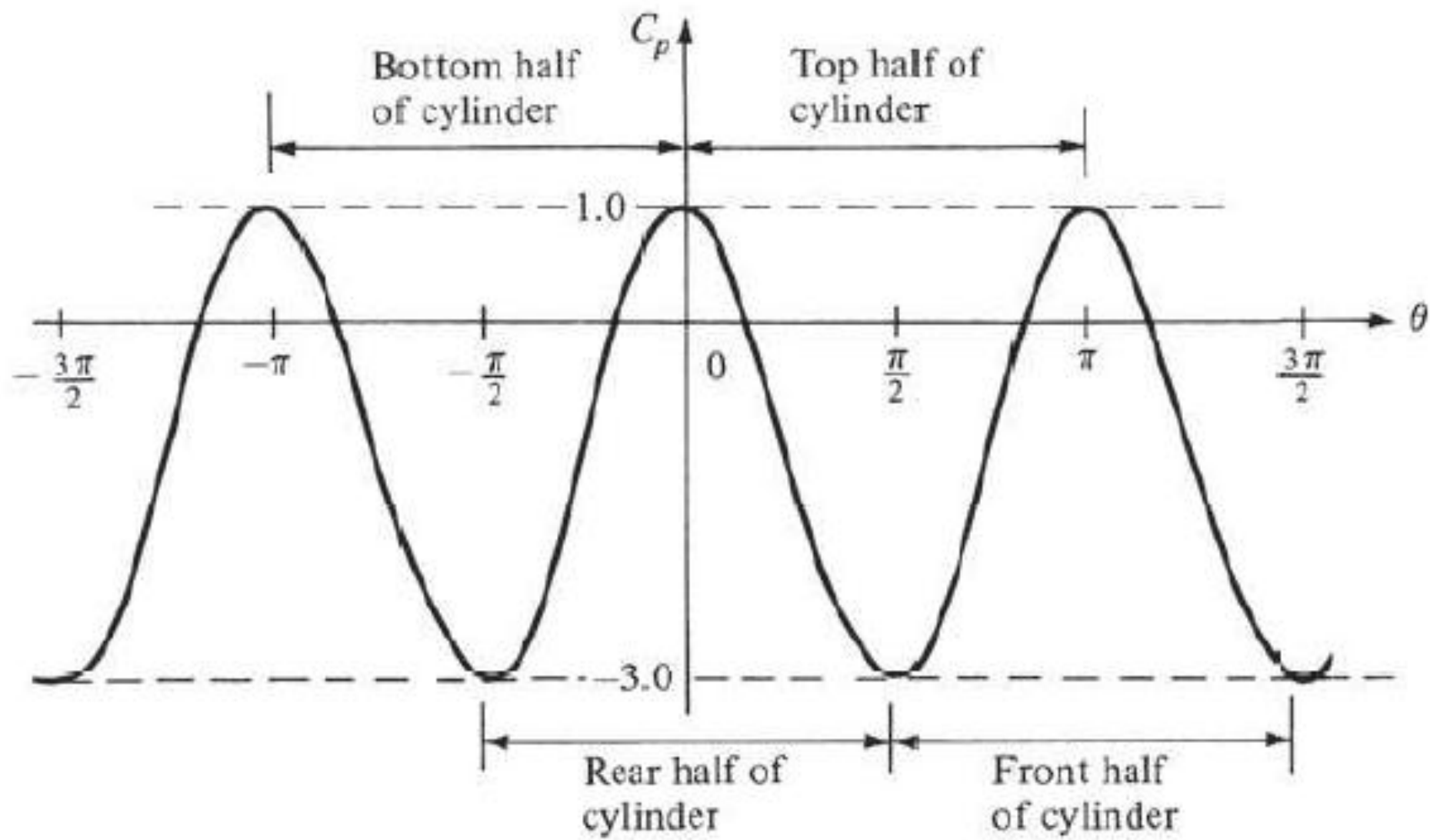


Figure 3.27 Sign convention for V_θ in polar coordinates.

$$C_p = 1 - 4 \sin^2 \theta$$





become, respectively,

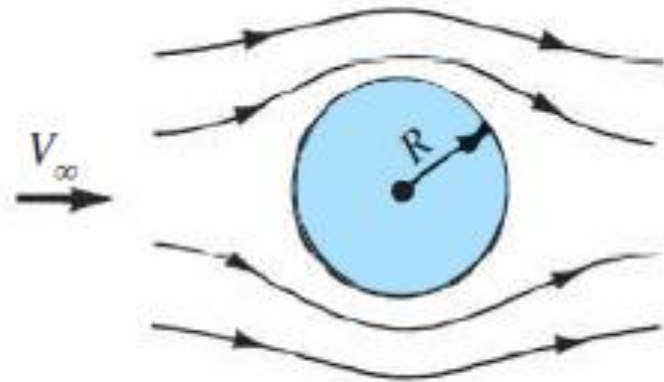
$$c_n = \frac{1}{c} \int_0^c (C_{p,l} - C_{p,u}) dx$$

$$c_a = \frac{1}{c} \int_{LE}^{TE} (C_{p,u} - C_{p,l}) dy$$

Type of flow	Velocity	ϕ	ψ
Uniform flow in x direction	$u = V_\infty$	$V_\infty x$	$V_\infty y$
Source	$V_r = \frac{\Lambda}{2\pi r}$	$\frac{\Lambda}{2\pi} \ln r$	$\frac{\Lambda}{2\pi} \theta$
Vortex	$V_\theta = -\frac{\Gamma}{2\pi r}$	$-\frac{\Gamma}{2\pi} \theta$	$\frac{\Gamma}{2\pi} \ln r$
Doublet	$V_r = -\frac{\kappa}{2\pi} \frac{\cos \theta}{r^2}$ $V_\theta = -\frac{\kappa}{2\pi} \frac{\sin \theta}{r^2}$	$\frac{\kappa}{2\pi} \frac{\cos \theta}{r}$	$-\frac{\kappa}{2\pi} \frac{\sin \theta}{r}$

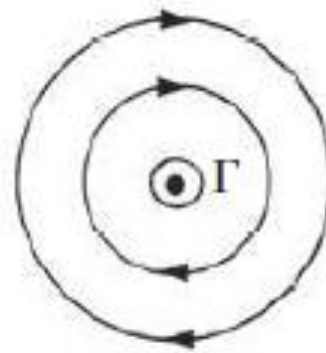
VORTEX FLOW: OUR FOURTH ELEMENTARY FLOW

LIFTING FLOW OVER A CYLINDER



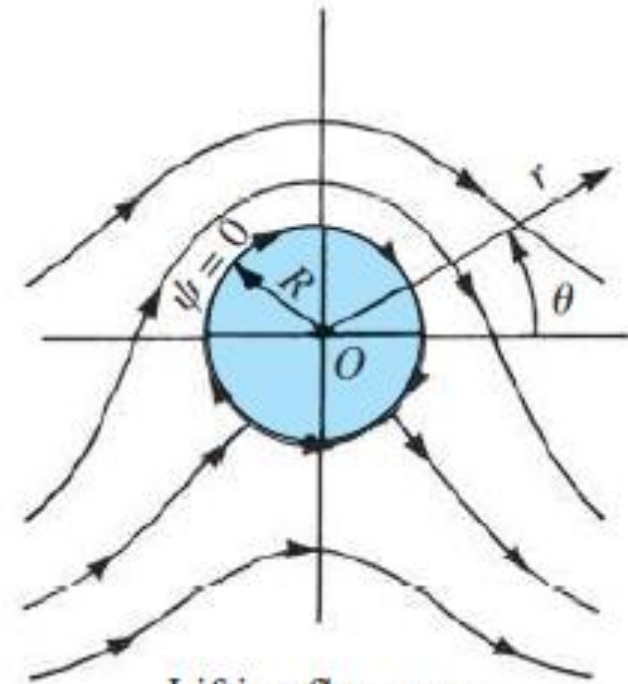
Nonlifting flow
over a cylinder

+



Vortex of
strength Γ

=



Lifting flow over
a cylinder

$$\psi_1 = (V_\infty r \sin \theta) \left(1 - \frac{R^2}{r^2} \right)$$

$$\psi_2 = \frac{\Gamma}{2\pi} \ln r + \text{const}$$

Since the value of the constant is arbitrary, let

$$\text{Const} = -\frac{\Gamma}{2\pi} \ln R$$

$$\psi_2 = \frac{\Gamma}{2\pi} \ln \frac{r}{R}$$

$$\psi = \psi_1 + \psi_2$$

$$\psi = (V_\infty r \sin \theta) \left(1 - \frac{R^2}{r^2} \right) + \frac{\Gamma}{2\pi} \ln \frac{r}{R}$$

Lifting flow over a cylinder of radius R ,

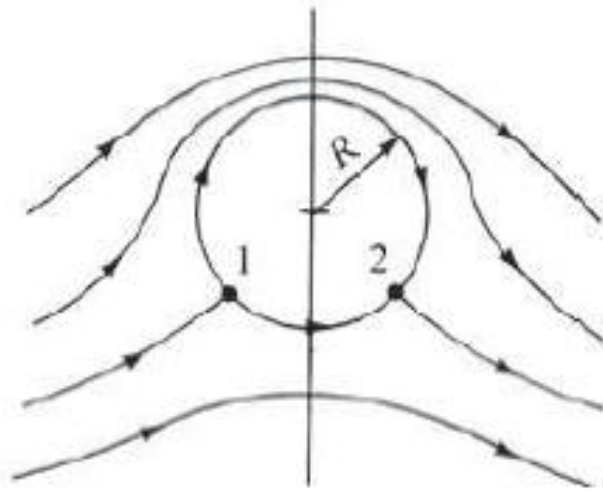
$$V_r = \left(1 - \frac{R^2}{r^2}\right) V_\infty \cos \theta$$

$$V_\theta = -\left(1 + \frac{R^2}{r^2}\right) V_\infty \sin \theta - \frac{\Gamma}{2\pi r}$$

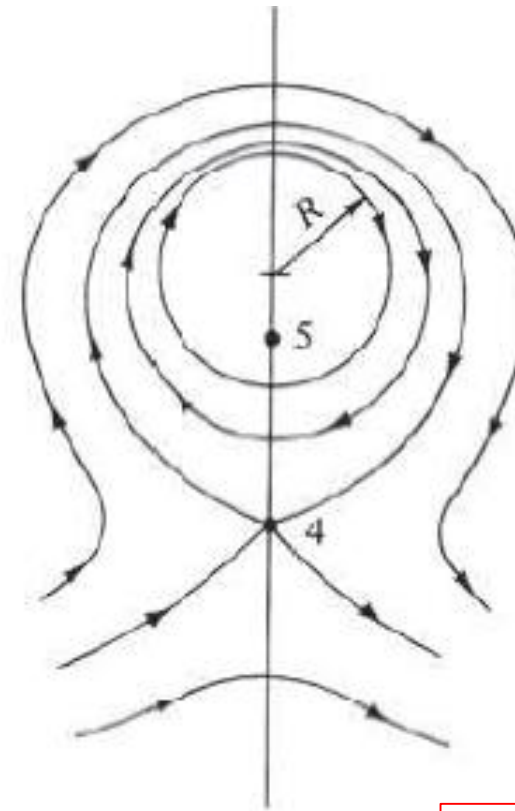
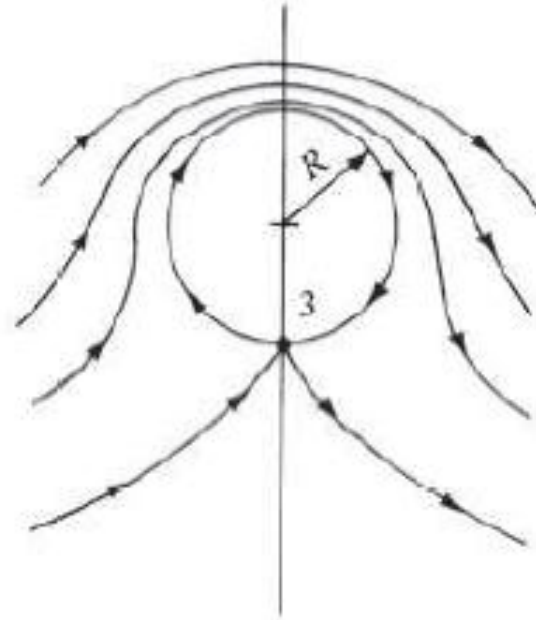
To locate the stagnation points in the flow, set $V_r = V_\theta = 0$ in Equations (3.119) and (3.120) and solve for the resulting coordinates (r, θ) :

$$V_r = \left(1 - \frac{R^2}{r^2}\right) V_\infty \cos \theta = 0$$

$$V_\theta = -\left(1 + \frac{R^2}{r^2}\right) V_\infty \sin \theta - \frac{\Gamma}{2\pi r} = 0$$



(a) $\Gamma < 4\pi V_\infty R$



(c) $\Gamma > 4\pi V_\infty R$

$$\theta = \arcsin \left(-\frac{\Gamma}{4\pi V_\infty R} \right)$$

$$r = \frac{\Gamma}{4\pi V_\infty} \pm \sqrt{\left(\frac{\Gamma}{4\pi V_\infty} \right)^2 - R^2}$$

satisfied by $r = R$; however, it is also satisfied by $\theta = \pi/2$ or $-\pi/2$. Substituting $\theta = -\pi/2$ into Equation (3.122), and solving for r , we have

The velocity on the surface of the cylinder is given by Equation (3.120) with $r = R$:

$$V = V_\theta = -2V_\infty \sin \theta - \frac{\Gamma}{2\pi R}$$

In turn, the pressure coefficient is obtained by substituting Equation (3.125) into Equation (3.38):

$$C_p = 1 - \left(\frac{V}{V_\infty} \right)^2 = 1 - \left(-2 \sin \theta - \frac{\Gamma}{2\pi R V_\infty} \right)^2$$

or

$$C_p = 1 - \left[4 \sin^2 \theta + \frac{2\Gamma \sin \theta}{\pi R V_\infty} + \left(\frac{\Gamma}{2\pi R V_\infty} \right)^2 \right] \quad (3.126)$$

$$\begin{aligned}
c_d &= c_a = \frac{1}{c} \int_{\text{LE}}^{\text{TE}} (C_{p,u} - C_{p,l}) dy \\
c_d &= \frac{1}{c} \int_{\text{LE}}^{\text{TE}} C_{p,u} dy - \frac{1}{c} \int_{\text{LE}}^{\text{TE}} C_{p,l} dy
\end{aligned} \tag{3.127}$$

Converting Equation (3.127) to polar coordinates, we note that

$$y = R \sin \theta \quad dy = R \cos \theta d\theta \tag{3.128}$$

Substituting Equation (3.128) into (3.127), and noting that $c = 2R$, we have

$$c_d = \frac{1}{2} \int_{\pi}^0 C_{p,u} \cos \theta d\theta - \frac{1}{2} \int_{\pi}^{2\pi} C_{p,l} \cos \theta d\theta \tag{3.129}$$

$$c_d = -\frac{1}{2} \int_0^\pi C_p \cos \theta \, d\theta - \frac{1}{2} \int_\pi^{2\pi} C_p \cos \theta \, d\theta$$

or

$$c_d = -\frac{1}{2} \int_0^{2\pi} C_p \cos \theta \, d\theta \quad (3.130)$$

Substituting Equation (3.126) into (3.130), and noting that

$$\int_0^{2\pi} \cos \theta \, d\theta = 0 \quad (3.131a)$$

$$\int_0^{2\pi} \sin^2 \theta \cos \theta \, d\theta = 0 \quad (3.131b)$$

$$\int_0^{2\pi} \sin \theta \cos \theta \, d\theta = 0 \quad (3.131c)$$

$$\int_0^{2\pi} \cos \theta \, d\theta = 0$$

$$\int_0^{2\pi} \sin^2 \theta \cos \theta \, d\theta = 0$$

$$\int_0^{2\pi} \sin \theta \cos \theta \, d\theta = 0$$

$$c_d = 0$$

The lift on the cylinder can be evaluated in a similar manner as follows. From Equation (1.15) with $c_f = 0$,

$$c_l = c_n = \frac{1}{c} \int_0^c C_{p,l} \, dx - \frac{1}{c} \int_0^c C_{p,u} \, dx \quad (3.133)$$

Converting to polar coordinates, we obtain

$$x = R \cos \theta \quad dx = -R \sin \theta \, d\theta \quad (3.134)$$

Substituting Equation (3.134) into (3.133), we have

$$c_l = -\frac{1}{2} \int_{\pi}^{2\pi} C_{p,l} \sin \theta \, d\theta + \frac{1}{2} \int_{\pi}^0 C_{p,u} \sin \theta \, d\theta \quad (3.135)$$

Again, noting that $C_{p,l}$ and $C_{p,u}$ are both given by the same analytic expression, namely, Equation (3.126), Equation (3.135) becomes

$$c_l = -\frac{1}{2} \int_0^{2\pi} C_p \sin \theta \, d\theta \quad (3.136)$$

Substituting Equation (3.126) into (3.136), and noting that

$$\int_0^{2\pi} \sin \theta \, d\theta = 0 \quad (3.137a)$$

$$\int_0^{2\pi} \sin^3 \theta \, d\theta = 0 \quad (3.137b)$$

$$\int_0^{2\pi} \sin^2 \theta \, d\theta = \pi \quad (3.137c)$$

we immediately obtain

$$c_l = \frac{\Gamma}{RV_\infty} \quad (3.138)$$

From the definition of c_l (see Section 1.5), the lift per unit span L' can be obtained from

$$L' = q_\infty S c_l = \frac{1}{2} \rho_\infty V_\infty^2 S c_l \quad (3.139)$$

we immediately obtain

$$c_l = \frac{\Gamma}{RV_\infty} \quad (3.138)$$

From the definition of c_l (see Section 1.5), the lift per unit span L' can be obtained from

$$L' = q_\infty S c_l = \frac{1}{2} \rho_\infty V_\infty^2 S c_l \quad (3.139)$$

Here, the planform area $S = 2R(1)$. Therefore, combining Equations (3.138) and (3.139), we have

$$L' = \frac{1}{2} \rho_\infty V_\infty^2 2R \frac{\Gamma}{RV_\infty}$$

or

$$\boxed{L' = \rho_\infty V_\infty \Gamma} \quad (3.140)$$

THANK

YOU

Q...??????

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